MUHS final-year BPTh (IV BPTh), 2012 syllabus

Community Physiotherapy

Short notes for revision: one sheet per topic, in syllabus order, with the page of the full notes for each.



Sheets	27 sheets: every topic in the syllabus, then the exam skills
Built from	The full notes and the 20 MUHS question papers, Summer 2016 to Summer 2026 (code 51414)
Content checked	21 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/community/

Educational notes for exam preparation. Not medical advice. Sources for every fact are listed in the full notes.

How to use the short notes

These sheets are for revision, after you have read a topic once in the full notes. Each topic has its own sheet of one to three pages, in the same order as the full notes.

1. **Read the yellow line first.** It says how the topic is asked in the MUHS paper.
2. **Cover a box and recall it,** then check. Bold words and numbers are the ones to write in the answer.
3. **Tick a square at the top right of a sheet** each time you revise it. Aim for day 1, day 7 and day 21.
4. **Need more?** Each sheet gives the page of the same topic in the full notes.

What the colours mean

Grey
Definitions, types, medical and surgical management

Orange
Disease process, clinical features and the problems they cause

Blue
Investigations, scales and normal values

Green
Physiotherapy: assessment, ICF, goals, treatment and doses

Amber
Precautions, contraindications, stop criteria and complications

Yellow
Exam plan: answer structure, what to draw, keywords

The paper (20 papers, 2016–2026): 80 marks in 3 hours. Short answers: any 5 of 6 at 3 marks and any 5 of 6 at 7 marks. Long answers: two questions, each any 1 of 2 at 15 marks. Most long cases are a person in a community setting (a worker in pain, a pregnant or postnatal woman, an older person, a person with disability) or a planning question; answers need ICF, goals, treatment with doses, and ergonomic, home or community measures.

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Health, disease and the health care delivery system in India

9 papers: WHO definition, the three tiers, PHC, scope of community physiotherapy.

Full notes p. 5

Asked: 9 of 20 papers, short answers only. **7 marks:** functioning of the PHC (S22); principles and scope of community physiotherapy (W25). **3 marks:** PHC setting (S17), WHO definition of health (S19), socioeconomic factors and PT services (S20, W24), delivery system in India (S21), role of PT (S24), levels of health care delivery (S25).

Health and prevention (3m S19)

WHO (1948): "Health is a state of **complete physical, mental and social well-being** and not merely the absence of disease or infirmity." Dimensions: physical, mental, social (+ spiritual). **Disease:** disturbance of structure or function of body or mind, with signs and symptoms. Health is a **continuum**: positive health → ill-health → death. **Determinants:** genes; lifestyle; environment; socioeconomic; health services.

Primordial	Stop risk factors arising: active schools
Primary	Healthy people: fitness, ergonomics, antenatal exercise, falls exercise
Secondary	Early diagnosis: school posture, developmental screening, early back care
Tertiary	Limit disability: stroke, amputee, cardiac rehab, CBR

Primary health care (3m S17)

Alma-Ata 1978: essential care, **universally accessible** through **full community participation**, at affordable cost; the **first level of contact**, close to where people live and work. **Principles:** equity; community participation; intersectoral coordination; appropriate technology; prevention and promotion. **Eight elements:** health education; food and nutrition; safe water and sanitation; mother and child health, family planning; immunisation; local endemic disease control; treatment of common illness and injury; essential drugs.

Three tiers in India (3m S21, S25)

Unit	Population (hilly, tribal); staff
Village	ASHA (about 1 per 1,000), anganwadi worker
Sub-centre	**5,000** (3,000); ANM, male health worker, community health officer
PHC	**30,000** (20,000); medical officer; **4–6 beds**; for about 6 sub-centres
CHC	**1,20,000** (80,000); **30 beds**; 4 specialists; first referral unit for 4 PHCs
District, tertiary	District hospital (PT, early intervention); medical colleges, AIIMS, national institutes

Ayushman Arogya Mandirs: upgraded sub-centres and PHCs, **12 service packages** (non-communicable disease screening, elderly, palliative).

Functions of the PHC (7m S22)

Medical care and referral; **mother and child** (antenatal, delivery, postnatal, newborn) and family planning; **immunisation**; TB, malaria, leprosy and non-communicable disease control; national programmes; health education and nutrition; water and sanitation; **vital statistics**; laboratory and essential drugs; **training ANMs, ASHAs**; school health; Arogya Mandir packages.

Role of PT by level (3m S24)

Primary: fitness, ergonomics, antenatal classes, screening, training ASHAs, home programmes, CBR. **Secondary:** CHC and district outpatients and wards (fractures, stroke, chest, post-op). **Tertiary:** ICU, cardiac, neuro, paediatric, sports.

Socioeconomic barriers (3m S20, W24)

Out-of-pocket cost, lost wages; PTs in **cities and private clinics**; distance; low awareness ("rest cures"), healers; gender (permission); floor-level homes, Indian toilets; PM-JAY mainly hospital care. **Solutions:** CBR, camps, tele-rehab, posts at CHCs and Arogya Mandirs.

Community PT: principles and scope (7m W25)

PT for the whole community, **preventive and promotive**, where people live and work. **Principles:** prevention first; **equity**; participation; ICF; teamwork, intersectoral; low-cost technology; self-management; evidence, sustainability. **Scope:** fitness (p. 4); women's health (p. 9); geriatrics (p. 21); industrial health (p. 31); CBR (p. 26); non-communicable diseases; disaster (p. 29); school, palliative care.

Exam plan

3m delivery system: tiers (1) → units with numbers (1.5) → Arogya Mandirs (0.5). **7m PHC:** population, staff → 10+ functions → PT place. **Draw:** three-tier pyramid with numbers. **Underline:** complete well-being, Alma-Ata 1978, 5,000 / 30,000 / 1,20,000, Arogya Mandir, prevention first, equity.

Physical fitness: components, evaluation and exercise prescription

12 papers: FITT, aerobic and anaerobic exercise, MET, field tests, bed rest.

Full notes p. 11

Asked: 12 of 20 papers, short answers only. **7 marks:** field tests in athlete training, principles and phases of exercise, bed rest deconditioning (all S16); energy systems (W16); hazards of a sedentary lifestyle (W18); aerobic and anaerobic exercise with the MET (W22); aerobic exercise and circuit training (S25); exercise in hypertension, principles of exercise prescription (W25). **3 marks:** bed rest (W16), diabetic precautions (S18), principles of aerobic training (S19), FITT (W20), aerobic exercise (S21), aerobic vs anaerobic (W24), adult fitness tests (S26).

Definitions and components

Physical activity: any movement by skeletal muscles that uses energy. **Exercise:** planned, structured, repeated, to improve or keep fitness. **Physical fitness:** attributes that relate to the ability to perform physical activity (Caspersen 1985). **Health-related (5):** cardiorespiratory endurance, strength, muscular endurance, flexibility, body composition. **Skill-related (6):** agility, balance, coordination, speed, power, reaction time.

Fitness tests (3m S26) and field tests (7m S16)

Tests: cardiorespiratory: **YMCA step**, Rockport walk, Cooper run, 6MWT; strength: **1RM**, grip; endurance: push-up, curl-up; flexibility: **sit-and-reach**, back scratch; body composition: BMI, waist, skinfolds; agility: Illinois, T-test; balance: stork, Y-balance; coordination: wall toss; speed: 30–50 m sprint; power: jumps; reaction: ruler-drop.

- **Field test:** away from the laboratory, no expensive kit (NSCA); must be **valid** and **reliable**.
- **Beep test:** 20 m shuttles to beeps, **8.5 km/h, +0.5 each minute**; last stage → VO_2max. **Yo-Yo IR:** 2 × 20 m, **10 s jog**; level 1 from 10 km/h; distance. **Cooper 12 min:** r 0.897 with treadmill.
- **RAST:** 6 × 35 m, 10 s rest; power = mass × distance^2 ÷ time^3. **Jumps:** Sargent (reach difference), broad jump.
- **T-test:** 10 yd forward, shuffle 5 left, 10 right, 5 back, backpedal (reliability 0.98). **Illinois** excellent (16–19 y): **< 15.2 s** boys, **< 17.0 s** girls. These test change of direction, not reactive agility.
- **Order (NSCA):** non-fatiguing (height, weight, flexibility, skinfold, vertical jump) → agility → strength, power → sprint → muscular endurance → anaerobic → **aerobic last**. Limits: weather, motivation; values predicted.

Energy systems (7m W16)

- **ATP** is the only direct fuel; stored for seconds. **Phosphagen:** creatine phosphate + ADP → ATP; fastest, smallest. **Fast glycolysis:** glucose → lactate, **net 2 ATP**. **Oxidative:** mitochondria, **~30–32 ATP**/glucose; slowest, largest.
- **NSCA: 0–6 s** phosphagen; 6–30 s + glycolysis; **30 s–2 min** glycolysis; 2–3 min glycolysis + oxidative; **> 3 min** oxidative.
- **Lactate threshold** 50–60% VO_2max untrained, 70–80% trained. **EPOC** after exercise.
- **Work : rest:** phosphagen **1:12–1:20**; glycolysis 1:3–1:5; oxidative 1:1–1:3 or continuous.

Aerobic vs anaerobic (3m S21, W24; 7m W22) and MET

	Aerobic	Anaerobic
Energy	Rhythmic, large muscles; oxygen; below lactate threshold	Short, intense; ATP-PC, glycolysis; above it
Length, by-product	Minutes to hours; CO_2, water	Seconds to ~2 min; **lactate**
Fibres, gains	Type I; ↑ VO_2max, ↓ resting HR, BP	Type II; strength, power, speed
Examples	Brisk walking, cycling, swimming	Sprinting, weightlifting, jumping

MET: rest, **3.5 ml O_2/kg/min** (~1 kcal/kg/h). Sedentary **≤ 1.5**; light 1.6–2.9; moderate **3.0–5.9**; vigorous **≥ 6**. Yoga 2.3, slow walk 3.0, **brisk walk 4.8**, cycling 6.8, running ~10 km/h 9.3. **kcal/min = METs × 3.5 × kg ÷ 200**. MET-min/week: ACSM **500–1000**. MET is an estimate: check HR, RPE.

Principles (3m S19; 7m S16) and FITT-VP, ACSM (3m W20)

Overload; **progression** (duration → frequency → intensity, ~10% a week); **specificity**; **reversibility** ("use it or lose it"); individuality; recovery (else overtraining).

	Aerobic	Resistance
F	**5+ days** moderate or **3+** vigorous	**2–3 days**
I	**40–59% HRR** (RPE 12–13); vigorous 60–89%	**60–70% 1RM**; 40–50% older
T	30–60 min (**150/week**) or 20–60 vigorous (75)	**2–4 × 8–12**
T, V, P	Walking, cycling; 500–1000 MET-min	Multi-joint; more load when easy

Flexibility: 2–3 days; hold **10–30 s** (30–60 older). **Neuromotor:** 2–3 days. **Karvonen:** resting HR + (max – resting) × intensity.

Prescription steps (7m W25) and phases (7m S16)

Screen (history, risk factors, drugs, clearance) → **test**, set goals → write **FITT-VP** → teach warm-up, cool-down, **talk test, RPE**, warning signs → **retest every 4–6 weeks**. **Phases: warm-up 5–10 min** → **conditioning 20–60 min** → **cool-down 5–10 min** (no pooling, dizziness) → **stretch ≥ 10 min**. Stages: initial → improvement → maintenance.

Circuit training (7m S25)

Stations one after another with little rest, muscle groups alternating, aerobic stations added; repeated. **Morgan and Adamson**, Leeds, 1953. **Method: 10–15 stations**; **30–40 s or 12–15 reps** at **40–60% 1RM**; **15–30 s** to move on; **1–3 circuits** (~30 min). **Types:** weight, aerobic, high-intensity (12 × 30 s, 10 s change), circuit class therapy, sport-specific. **Effect:** VO_2max +5% men, +8% women; **+11%** with HR at 70–80% max. Stroke circuit class: 6MWT +61 m (Cochrane 2017). **Precautions:** technique; no breath-holding; RPE.

Sedentary lifestyle (7m W18)

Sedentary behaviour: awake, **≤ 1.5 METs**, sitting, reclining or lying. **Physical inactivity:** not meeting guidelines (SBRN 2017); separate risks. **31%** of adults worldwide, India **~49%** (women 57%) insufficiently active (2022); ICMR-INDIAB **54.4%** inactive. **20–30% higher death risk** (WHO). High sitting: death HR 1.24; **diabetes 1.91**; CVD death 1.18; cancer death 1.17; depression RR 1.25. Also obesity, colon and breast cancer, sarcopenia, osteoporosis, back pain. Risk rises above **6–8 h/day** sitting; **60–75 min/day** moderate activity offsets it.

Prescribe: WHO **150–300 min** + strength 2 days; sit less; **2-min walk every 20 min** (glucose −24–30%); desk workers **2 h → 4 h/day** standing and light walking; sit-stand desk; stairs, walking meetings, reminders.

Bed rest deconditioning (7m S16; 3m W16)

Heart: ↑ resting HR, ↓ stroke volume, ↓ VO_2max, postural hypotension, DVT. **Lungs:** basal collapse, pneumonia. **Muscle:** antigravity wasting, **~1–1.5% strength a day**. **Bone, joints:** bone loss, urine calcium, contractures. **Skin:** sores. **Metabolic:** insulin resistance. **Gut, kidney:** constipation, stones. **Mind:** confusion, low mood. **Prevent:** sit out, walk early; breathing; ankle pumps; resisted exercise; turning; early weight-bearing.

Exercise in hypertension (7m W25)

India: **35.5%** of adults (ICMR-INDIAB 2023). **During:** systolic rises, diastolic level. **Acute:** BP lower **up to 22 h** (post-exercise hypotension). **Training: 5–8 mmHg** fall in hypertension. Edwards 2023: aerobic 4.5/2.5, resistance 4.6/3.0, combined 6.0/2.5, **isometric 8.2/4.0** (best). **How:** noradrenaline −29%, renin −20%, resistance −7%; endothelium; stiffness; weight, insulin. **ACSM:** aerobic **5–7 days**, 40–59% HRR, **≥ 30 min**; resistance **2–3 days**, 60–70% 1RM, 2–4 × 8–12. Classic "no weights" is outdated: no breath-holding.

Precautions: diabetes (3m S18), hypertension

- **Diabetes (ADA 2016): ≥ 150 min/week**, ≥ 3 days, **≤ 2 days in a row off**; resistance 2–3 days. Glucose before and after if on insulin or sulphonylureas; fast sugar; not the exercising limb; lows **6–15 h later**, risk to **48 h**; no exercise with high glucose + ketones.
- Feet: check, footwear, cycling or swimming if ulcer. Retinopathy: no heavy lifting, jarring, head-down, breath-holding. Autonomic neuropathy: RPE, slow position change.
- **Hypertension:** no start if resting BP **> 200/110**; stop test **> 250/115**; long warm-up, cool-down (vasodilators, alpha-blockers); RPE with beta-blockers; diuretics: dehydration, low potassium.

Exam plan

7m prescription: definition (1) → principles (2) → FITT-VP with numbers (3) → safety (1). **7m field tests:** definition, valid, reliable → test table by component → NSCA order → pros and limits.

Draw: energy-system time line; circuit of 8 stations; session HR curve. **Underline:** ATP–PC, 1 MET = 3.5 ml/kg/min, ≤ 1.5 METs, overload, reversibility, FITT-VP, heart rate reserve, post-exercise hypotension, hypoglycaemia.

Revised ☐☐☐

Obesity in children and adults: body composition, BMI and weight management

14 papers, with 10 long cases on weight reduction.

Full notes p. 26

Asked: 14 of 20 papers, 10 long cases: bride crash dieting (W22, S25), banker after caesarean (S21), housewife (S19), lecturer with diabetes and BP (S20, W24), school screening (W20), boy of 14 (W21), knee pain (S16, W16). **7 marks:** body composition methods (S16, W16), set-point theory (S20), causes of childhood obesity (S20, W21, W24), effects on adult health (W20), grades and significance (S24), anthropometry (S26). **3 marks:** definition and types (W16, S25), grades (W17), BMI and uses (W22, S23), fat mass and fat-free mass (S24).

Definition and types (3m W16, S25)

Obesity (WHO): abnormal or excessive fat accumulation that may impair health; screen by BMI and waist, confirm by body fat. **BMI = weight (kg) ÷ height (m)²**: 68 ÷ 1.60^2 = 68 ÷ 2.56 = 26.6.

- **Distribution: android** (apple, central; men; more diabetes and heart disease) vs **gynoid** (pear, hips and thighs; women).
- **Fat cells: hyperplastic** (more cells; childhood; harder to reverse) vs **hypertrophic** (bigger cells; adult onset).
- **Cause: primary** (intake more than use) vs **secondary** (hypothyroidism, Cushing's, PCOS, steroids, Prader–Willi). **Severity:** BMI grade.

Grades (3m W17; 7m S24), BMI uses (3m W22, S23)

	WHO	Asian Indian 2009
Normal	18.5–24.9	**18.5–22.9**
Overweight	25–29.9	**23–24.9**
Obese	I 30–34.9; II 35–39.9; III 40+	**25 or more**

Indian 2025: any BMI **over 23** = obesity (no overweight step). Stage 1: no organ or activity effect. Stage 2: + waist **≥ 90 cm men, ≥ 80 cm women** or waist-to-height **over 0.5**, + activity limit or related disease. Grades I 23–24.9, II 25–27.5, III 27.6–32.4, IV 32.5+. Body fat obesity: over **25.5% men, 38% women**.

Significance: risk rises each grade; Asian Indians have more (abdominal) fat at the same BMI → lower cut-offs; grade guides lifestyle → drugs → surgery; waist catches risk at normal BMI. **BMI uses:** cheap screening, tracking, treatment decisions, surveys. **Limits:** fat vs muscle (athlete), no distribution, age charts in children.

Body composition (7m S16, W16; 3m S24), anthropometry (7m S26)

Fat mass: essential (nerves, marrow, organs; more in women) + storage fat. **Fat-free mass:** muscle, bone, organs, water (~three-quarters water); sets **resting metabolic rate** → keep it (resistance work, protein, no crash diets). Body fat % = fat mass ÷ weight × 100.

Tape and scale	BMI, waist, hip; WHR risk **≥ 0.90 men, ≥ 0.85 women**; waist-to-height over 0.5
Skinfolds	Calipers (biceps, triceps, subscapular, suprailiac) → % fat; cheap; poor in severe obesity
Bioimpedance	Current faster through water-rich lean tissue; hydration, food, exercise alter it
Underwater	**Archimedes**: fat floats; old reference; full breath-out
Bod Pod, DEXA	Air displacement; two X-ray energies (fat, lean, bone); costly. CT, MRI: visceral fat

Anthropometry: weight (light clothes) and height (stadiometer); **waist midway between lowest rib and iliac crest** at end of normal breath out; hip at widest buttocks; ratios; skinfolds; mid-arm and **neck** (sleep apnoea). Same time, same side, mean of two; monthly.

Causes, set point (7m S20), adult effects (7m W20)

Causes: high-calorie food, sugary drinks; sedentary work, screens; poor sleep, stress; genes, family habits; after pregnancy or caesarean; menopause; steroids; hypothyroidism, PCOS; stopping sport or smoking; ageing.

Set-point theory: hypothalamus (with **leptin** from fat) defends a "set" weight like a **thermostat**; dieting → hunger up, metabolism down → regain after crash diets. Slow loss with exercise and lasting habits may settle a lower set point.

Effects: metabolic (type 2 diabetes, metabolic syndrome, fatty liver, gallstones, gout); **heart** (hypertension, CHD, failure, stroke, DVT); **lungs** (effort breathlessness, sleep apnoea, hypoventilation); **joints** (knee and hip OA, back pain, flat feet, heel pain); **women** (PCOS, infertility, gestational diabetes, pre-eclampsia, stress incontinence); **cancer** (breast, colon, endometrial); **mind** (depression, stigma, shorter life).

Childhood obesity (7m S20, W21, W24; LAQ W20, W21)

Measure: BMI-for-age on **IAP 2015** charts (5–18 y): overweight from the **adult-equivalent 23** line, obesity from **27**; WHO standards under 5. **Causes:** fast food, sugary drinks, portions; little play; **screen time**; short sleep; obese parents, family habits; no safe play space; academic pressure; emotional eating; rarely endocrine or genetic; birth weight, mother's diabetes, formula. **Ill effects:** early type 2 diabetes, BP, fatty liver, sleep apnoea; flat feet, knock knees, **slipped capital femoral epiphysis**; early puberty, PCOS; teasing; persists into adult life.

School screening and promotion (W20)

Khelo India (9–18 y): BMI (+ waist); **600 m run or walk**; 50 m dash; partial curl-ups; push-ups (modified for girls); sit and reach. Consent → start-of-year test by teachers, PTs → report card → refer obesity, high BP, low fitness → retest at year end.

Promotion: ≥ 60 min/day moderate to vigorous, muscle-bone work 3 days a week (WHO); daily PE, active travel; no junk food or sugary drinks in canteens, home tiffin, water; limit screens; sleep; **whole family** changes; no blame, praise effort not weight; fun non-competitive games.

ICF (S25 worked case: 27, 5 ft 3 in, 68 kg, marriage in 3

months)

Status: BMI **26.6** (WHO overweight; Asian obese; 2025 grade II); waist 99 cm, waist-to-height **0.62**; fat **50%**; sit and reach **–3 cm** poor; 12 modified push-ups fair (good 15–20); 34 squats baseline; test aerobic (6MWT, YMCA step).

- **Structure/function:** central fat, poor flexibility, fair endurance; crash-diet fatigue; rule out thyroid, PCOS.
- **Activity:** low exercise tolerance; tires in busy wedding days.
- **Participation:** anxious about appearance; social events.
- **Environment:** wedding food, social pressure, little time (barriers); clinic, family (facilitators). **Personal:** motivated, short deadline, believes in crash dieting.

Goals: short-term (4 weeks): stops crash diet, dietitian plan (~500 kcal deficit); **0.5 kg a week**; 150 min/week; sit and reach 0 cm. Long-term (3 months): ~5 kg (7%), waist –5 cm; **over 250 min/week**; 15–20 push-ups; healthy eating after marriage.

Weight management: FITT and lifestyle

- **Aerobic: ≥ 5 days**; **40–59% HRR** (talk, not sing) → vigorous, intervals; **30 → 60 min/day**; 150/week, then **over 250** for real loss and to keep it off (ACSM 2009); walking, cycling, water exercise, dance.
- **Resistance: 2–3 non-consecutive days**; 8–10 exercises, **2–3 × 8–12**; squats, lunges, bridges, modified push-ups, band rows, planks; keeps fat-free mass. **Flexibility daily:** 30 s × 2–3. Warm-up and cool-down 5–10 min.
- **Karvonen:** age 27, resting HR 80 → max 193, reserve 113 → 40–59% ≈ **125–147 beats/min**.
- **Diet** (dietitian): **500 kcal/day deficit ≈ 0.5 kg/week**; less refined carbs, sugar, fried food; more vegetables, pulses, protein. **Target 5–10% in 3–6 months**, then maintenance.
- **Lifestyle:** rising step target, stairs, less sitting, sleep 7–8 h, stress, food diary, family; review every 2–4 weeks.
- **Other cases:** housewife: home exercise around housework; banker (BMI 40): low impact, diastasis, desk set-up, breaks every 30–60 min; lecturer: sugar, BP, feet; boy: IAP charts, family.

Crash dieting and precautions

Crash diet fails: loses water and muscle, not fat; metabolism slows (set point); hunger rises; fatigue, dizziness, hair loss, irregular periods, gallstones, deficiency, disordered eating; **yo-yo** regain with more fat. **Precautions:** screen heart, diabetes, BP; low impact for painful knees; avoid heat; footwear; blood sugar in diabetes.

Obesity with knee pain (LAQ S16, W16)

Extra load, weak quads, inactivity, fat-tissue inflammation → early OA. **Exercise:** static quads, straight-leg raise, terminal extension, mini-squats in pain limits; hip abductors, extensors; hamstring, calf stretch; **static cycling, water walking** → walking. Loss of **over 5% in ~20 weeks** improved disability (Christensen 2007). **Joint protection:** no floor squatting or cross-legged sitting in flares; higher chair, western toilet; rail, up with good leg, down with painful; stick in opposite hand; cushioned shoes. Heat, ice, TENS as a bridge.

Exam plan (long case 15)

1. **Assessment (5):** BMI with working, waist, fat, fitness tests, risk screen; ICF, one-line diagnosis.
2. **Goals (3):** short and long term, measurable (kg, cm, minutes, test scores).
3. **Programme (7):** FITT with HR target; dietitian; lifestyle; precautions; review.

Draw: apple and pear with waist and hip sites; ICF model. **Underline:** Asian Indian cut-offs, waist 90/80 cm, waist-to-height 0.5, fat-free mass, set point, HRR, over 250 min a week, 5–10%, no crash diets.

Women in India: social issues, legal rights and health promotion

5 papers: social issues, legal rights, rural women. Full notes p. 37

Asked: 5 of 20 papers. **Long answers:** social issues and legal rights and benefits (W19, 7 + 8); approaching a rural community to teach primigravidae about PT, and management (S22, 5 + 10). **7 marks:** health promotion for rural women (S21); social issues affecting women's physical function (W21, W24).

Social issues and the body (7m W21, W24; W19)

Numbers: anaemia **57%** of women 15–49 (NFHS-5); maternal deaths **93** per 100,000 (2019–21); **1 in 4** married before 18; spousal violence **29%**; female literacy **65%** vs 82% (Census 2011).

Son preference	Fed less, less care: stunting, anaemia, weak bones
Early marriage, repeated pregnancy	Obstructed labour, fistula, **prolapse, incontinence**, maternal deaths
Poverty	Anaemia, fatigue, **osteoporosis**
Heavy work, double burden	**Back and neck pain** (squatting, **head-loading** water, fuel); knee OA; no time to exercise
Chulha smoke	Bronchitis, **COPD**, eye problems
No toilet	Holding urine and stool, urine infection, **pelvic floor problems**, unsafe night trips
Low literacy; restricted mobility	Late care; need permission or escort, **cannot attend repeated PT**
Violence, dowry	Injury, chronic pain, depression
Limits on exercise, sun; stigma	Obesity, **vitamin D deficiency**; infections, isolated widows

PT role: education in local language, safe carrying, pelvic floor training, antenatal and postnatal exercise, home visits, women therapists, schemes.

Legal rights and benefits (W19, 8 marks)

Constitution: Art. 14 equality; Art. 15 no sex discrimination; Art. 39 equal pay; Art. 42 maternity relief; **one-third panchayat seats** (73rd Amendment 1992); Lok Sabha too (2023 Act).

Acts: Maternity Benefit 1961 (2017): 26 weeks paid leave, nursing breaks, crèche; **Medical Termination of Pregnancy 1971 (2021)**: to 20 weeks, 24 special cases; **PCPNDT 1994**: no fetal sex determination; Dowry Prohibition 1961; **Domestic Violence 2005**: protection orders, shared home; Child Marriage 2006 (girl under 18); **workplace harassment 2013**: internal committees; Hindu Succession 2005: equal property; Code on Wages 2019.

Schemes: Janani Suraksha Yojana (JSY) 2005 (cash for facility delivery); **Janani Shishu Suraksha Karyakram (JSSK)** 2011 (free delivery, caesarean, drugs, transport); **Matru Vandana (PMMVY)** 2017 (₹5,000, first child; second if a girl); **Surakshit Matritva Abhiyan (PMSMA)** 2016 (free check-up on the **9th** monthly); Anaemia Mukt Bharat; **Ujjwala** (LPG); Beti Bachao Beti Padhao; Mission Shakti (helpline 181); Swachh Bharat (toilets).

Health promotion for rural women (7m S21)

1. **Know the community:** sarpanch, ASHA, ANM, anganwadi worker, self-help groups; problems; women's daily timetable.
2. **Existing platforms:** anganwadi, Village Health, Sanitation and Nutrition Day, group meetings, mahila mandals, PHC.
3. **Message:** iron, antenatal and postnatal exercise, pelvic floor, back care (water pots), bones, walking, clean fuel, toilets, screening.
4. **Methods:** local language, pictures, **demonstration and practice**, songs, short sessions, women educators, home visits.
5. Involve men, **mothers-in-law**, elders; link to schemes; **train local women**; evaluate (knowledge before and after, behaviour change).

Primigravidae in a village (LAQ S22, 5 + 10)

Approach (5): permission (PHC officer, panchayat, ASHA, ANM); list primigravidae from registers; sessions at anganwadi, **PMSMA clinic on the 9th**; invite husbands, mothers-in-law; flip charts, demonstrations.

Management (10): 1 posture, back care (lifting pots, side-lying rest); 2 exercise: **walking 30 min most days**, pelvic tilts, cat–camel, ankle exercises, warning signs; 3 pelvic floor (p. 14); 4 labour (p. 11): breathing, positions, JSY, JSSK, ambulance 108; 5 after birth: pelvic floor, abdominals, feeding posture, spacing. Throughout: iron, visits, danger signs. ASHAs repeat at home; refer pain, leakage, high BP, bleeding; evaluate knowledge, institutional delivery.

Exam plan

7m social issues: 2–3 numbers (1) → 8 issues each with its body effect (5) → PT role (1). **W19:** issues (7) + Constitution, Acts, schemes (8). **Draw:** no toilet → holding urine → infection, pelvic floor problems → training. **Underline:** anaemia 57%, head-loading, chulha, 26 weeks, PCPNDT, JSY, PMSMA, local language.

Pregnancy: physiological changes, discomforts and antenatal care

13 papers: changes in pregnancy, diastasis recti, exercise guidelines. Full notes p. 43

Asked: 13 of 20 papers. **Long answers:** nurse in the third trimester with heel and knee pain (W17, 5 + 5 + 5); changes and five discomforts with PT (S22, 4 + 5 + 6). **7 marks:** diastasis recti (W16, S22, S24), exercise guidelines (W18), physiological changes (S20, W20), aims and principles of obstetric PT (W21, W24), antenatal programme and class (S25). **3 marks:** diastasis (S17), contraindicated exercises (S17, S22), six changes (W17), early bird (W18), hyperemesis (W19), musculoskeletal dysfunctions (S21), diastasis assessment (W24).

Physiological changes (7m S20, W20; 3m W17)

Hormones	Progesterone, oestrogen, **relaxin** → ligament laxity; relaxed smooth muscle (veins, bowel, ureters)
Heart, blood	Blood volume **up 40–50%**, output up, resting HR **up 10–20**, BP dips mid-pregnancy; dilutional anaemia; clots easily → HR a poor guide, use **talk test** or Borg
Supine	From **~20 weeks** uterus presses the **inferior vena cava** → **supine hypotension**: side-lying or wedge under the right hip
Breathing	Diaphragm up, ribs widen; tidal and minute ventilation up; FRC down; mild effort breathlessness normal
Musculoskeletal	Weight; **centre of gravity forward**; lax pelvis and spine; abdominals stretch, separate; pelvic floor load; feet flatten → back, pelvic pain, leakage, balance
Kidney, gut	Frequency, stress leakage; nausea, heartburn, constipation, haemorrhoids
Metabolism, fluid	Metabolic rate and heat up, later insulin resistance → avoid overheating; fluid retention → ankle swelling, **carpal tunnel**, varicose veins

Weight gain (Institute of Medicine 2009): underweight 12.5–18 kg; **normal 11.5–16 kg**; overweight 7–11.5; obese 5–9.

Dysfunctions, discomforts and PT (3m S21; S22)

- **Low back pain** (posture, weight, laxity, weak abdominals): body mechanics, **pelvic tilts, cat–camel** in four-point, back and hip strengthening, heat not on abdomen, side-lying with pillow between knees, belt.
- **Pelvic girdle pain** (sacroiliac, symphysis): pelvic belt, **knees together** turning in bed and out of a car, no standing on one leg (sit to dress), short steps, stabilising exercise, crutches if severe.
- **Carpal tunnel:** night splint in neutral, elevation, nerve and tendon gliding, hand pumping. **Cramps:** calf stretch before bed, ankle pumps; pull toes up in a cramp.
- **Varicose veins, swelling:** elevation, ankle pumps, stockings, no long standing, walking, swimming. **Heel, foot, knee:** footwear, insoles, calf and plantar fascia stretch, strengthening, ice.
- **Breathlessness, rib pain:** upright, side-flexion stretch, lateral costal breathing. **Frequency, leakage:** daily pelvic floor training, the knack. **Constipation, heartburn:** walking, fibre, footstool; small meals, head-up sleep.

Hyperemesis gravidarum (3m W19)

Severe nausea and vomiting with **dehydration, weight loss, electrolyte imbalance**, often admitted; fluids, anti-emetics. **PT:** relaxation, slow breathing; **P6 acupressure bands** (some find helpful); bed and chair exercise, **ankle pumps**; early walking; energy conservation; support.

Diastasis recti: definition and test (3m S17, W24)

Separation of the two recti along the **linea alba** from stretching in pregnancy. First-time mothers: **33% at 21 weeks, 60% at 6 weeks after birth**, 45% at 6 months, 33% at 12 months (Sperstad 2016). Risks listed: multiple or large babies, many pregnancies, weak abdominals, obesity, lifting (study found none clear). Effects: doming, poor trunk support, back and pelvic pain, cosmetic worry; rarely hernia.

Test: crook lying, slow head and shoulder lift till scapulae clear the bed; fingers across midline **at the umbilicus, 4.5 cm above and 4.5 cm below**; **2 finger-widths or more** = diastasis (or calipers, ultrasound); note depth, tension, doming; pain, pelvic floor, function.

Diastasis recti: management (7m W16, S22, S24)

Pregnancy: roll to the side to get up; no straining or breath-holding; transversus + pelvic floor activation; posture; avoid doming exercises. **After birth: transversus drawing-in with pelvic floor**, pelvic tilts, side-lying and four-point → head lifts and curl-ups **only when the abdomen stays flat** → loaded functional training. Splint with crossed hands or binder for coughing and getting up; correct lifting, constipation care, weight; surgery only if large gap with hernia persists after long rehab.

Exercise guidelines (7m W18)

- **Benefits:** less gestational diabetes, pre-eclampsia, excess weight; better mood, sleep, back; **not linked with miscarriage or preterm birth** (Canadian 2018).
- **Dose: ≥ 150 min/week** moderate over **≥ 3 days**, ideally daily, aerobic + resistance (Canadian, ACOG); active women continue usual activity. **Intensity: talk test** (talk, not sing) or **Borg 12–14**.
- **Choose:** walking, stationary cycling, swimming, water exercise, low-impact aerobics, prenatal yoga and Pilates (not hot), light resistance, daily pelvic floor.
- **Precautions:** warm-up, cool-down; fluids; no heat; no long supine after first trimester; no breath-holding, straining, end-range or bouncing stretch; eat enough.

Contraindications and stop (3m S17, S22)

Avoid (ACOG 2020): contact sports (boxing, hockey, kabaddi); high fall risk (riding, skiing, gymnastics, off-road cycling); scuba, sky diving; **hot yoga**; unaccustomed altitude. Indian texts add: **double leg raises, full sit-ups**, long supine work, deep squats with pelvic pain, ballistic stretch, heavy lifting with breath-holding, jumping late.

Absolute: significant heart disease; restrictive lung disease; incompetent cervix or cerclage; multiple pregnancy at preterm risk; persistent 2nd or 3rd trimester bleeding; **placenta praevia after 26 weeks**; preterm labour; ruptured membranes; **pre-eclampsia**; severe anaemia. **Stop:** bleeding, painful contractions, leaking fluid, breathless before starting, dizziness, headache, chest pain, weakness, calf pain or swelling.

Obstetric PT: aims and principles (7m W21, W24)

Aims: adapt to changes, prevent and treat discomforts; safe fitness, weight control; protect back, pelvis, abdomen, pelvic floor; **prepare for labour** (knowledge, positions, relaxation, breathing, less fear); recovery after birth; prevent incontinence, prolapse, back pain, diastasis. **Principles:** assess, screen red flags, work with the obstetrician; moderate regular exercise by trimester; self-management; family, group support; **individualise** high-risk pregnancies.

Early bird (3m W18) and antenatal classes (7m S25)

Early bird (first trimester): changes, nutrition, back care, safe exercise, pelvic floor, warning signs, early discomforts; healthy habits early.

Antenatal course: from **~28 weeks, weekly, 6–8 sessions, groups of 8–12**, partner at some. **Goals:** knowledge, fitness, discomfort relief, pelvic floor, labour relaxation, breathing, positions, postnatal care, confidence. **Session (~90 min):** welcome and questions 10 → talk 20 → exercise 30 (warm-up, low-impact aerobic, strength, pelvic floor, stretch) → relaxation and breathing 20 → summary, handout 10. **Topics:** posture; pelvic floor; signs and stages of labour; positions and breathing (partner); caesarean; feeding, baby care.

ICF (W17: 25, nurse, third trimester, +20 kg, heel and knee pain)

Reasoning (5): extra weight loads heels and knees; **relaxin** → arches drop, feet roll in → plantar fascia strain, knees turn in; centre of gravity forward → leans back, locks knees; swelling; long standing on hard floors, lifting patients. 20 kg is above the 11.5–16 advised.

- **Structure/function:** heel pain on first steps, knee pain, flat pronated feet, tight calves, swelling, weak hip and thigh.
- **Activity:** long standing and walking, stairs, turning patients, rising from low seats.
- **Participation:** full shifts till her due date; housework.
- **Environment:** hard floors, long shifts (barriers); lighter duties, hoists (facilitators). **Personal:** motivated, health knowledge.

Management: cushioned arch-support shoes or insoles; calf and plantar fascia stretch **30 s × 3, 3 times a day**; cold-bottle rolling; low-dye taping; static quads, straight-leg raise, side-lying hip abduction **10 × 2–3**, no deep squats; **ankle pumps 20 every hour**, side-lying elevation, stockings; work: sitting duties, stool, anti-fatigue mat, **hourly breaks, no manual patient lifting**; water walking, cycling; dietitian (no dieting); pelvic floor, breathing, labour positions. **Report:** sudden face or hand swelling, headache, blurred vision (pre-eclampsia), bleeding, leaking, fewer baby movements, calf pain.

Exam plan

7m changes: hormones (1) → system table (5) → PT meaning (1). **7m diastasis:** definition → 3-level test → pregnancy and postnatal exercise → splinting. **Long case:** reasoning → ICF, one-line diagnosis → problem-wise PT with doses → warning signs.

Draw: diastasis test at three levels; pregnant posture (centre of gravity forward, knees locked); side-lying rest with pillows. **Underline:** relaxin, supine hypotension, two finger-widths, 150 min a week, talk test, placenta praevia, warning signs.

Labour and postnatal care

14 papers, with 9 long cases on labour, caesarean section and postnatal care.

Full notes p. 52

Asked: 14 of 20 papers, 9 long answers: episiotomy (S16), elective caesarean (S17), changes of labour and puerperium (S18), back pain 2 months after delivery (W18), executive after second baby (W20), forceps with dragging back pain (W21), postnatal complications (S23, 7 + 8), second stage (S24), back pain and heavy breasts (S26). **7 marks:** pain relief in labour (S17), breathing in labour (S19), PT during labour (S23). **3 marks:** caesarean complications (S17), positions in labour (W19), feeding ergonomics (S21), postpartum blues (S22), breast engorgement (S24), goals of postnatal care (W25).

Stages and changes of labour (S18)

First: cervix to **10 cm** (active from **5 cm**; up to 12 h first labour, 10 h later, WHO 2018): walking, upright, breathing, relaxation, massage, TENS. **Second:** full dilatation → birth: her position, push with the urge. **Third:** baby → placenta. **Fourth and puerperium:** first hours, then 6 weeks: breathing, ankle pumps, pelvic floor, feeding posture.

Changes: stronger, more frequent contractions; effacement, dilatation, descent, rotation; each contraction returns blood → output and BP rise; pain → over-breathing; slow stomach, fatigue; bladder pressed; perineum stretches greatly in the second stage.

PT in labour and positions (7m S23; 3m W19; S24)

Aims: less fear and pain, control, mobility, progress, save energy, protect perineum; taught in antenatal classes (p. 9).

First stage: walking, standing leaning forward, birthing ball, kneeling over a ball, **all-fours** (backache), side-lying for rest or epidural. Upright and mobile: first stage **~1 h 22 min shorter**, fewer caesareans and epidurals (Lawrence 2013). **Second stage: position of her choice** (kneeling, supported squat, all-fours, side-lying); **avoid lying flat** (great vessels, narrow outlet; WHO 2018).

Breathing in labour (7m S19)

- **Early:** slow relaxed breathing: sigh out at start, in by nose, out by mouth, sigh at end.
- **Strong contractions:** lighter, slightly faster at the peak, back to slow as it fades.
- **Transition: "pant-pant-blow"** to avoid pushing before full dilatation.
- **Second stage:** push with the urge, breathing out or short holds, **not long forced breath-holding**; pant as the head crowns.
- **Over-breathing** (tingling, dizzy): slow down, breathe into cupped hands. **Why:** focus, relaxation, oxygen for both, no exhausting holds.

Pain relief in labour (7m S17)

- **Relaxation: Mitchell's** or progressive, practised before; music. **Massage, warmth:** firm sacral pressure, warm packs (both WHO-recommended if she wants).
- **TENS:** two pairs on the back, **T10–L1** and **S2–S4**; low background + **boost button** in contractions; start early; not in water. Little effect on pain scores but most would use again, no harms (Dowswell 2009).
- Warm shower or bath; **continuous companion** (less fear, fewer drugs); medical: epidural, Entonox, opioids → PT adjusts positions.

Puerperium (S18) and goals (3m W25)

6 weeks. Involution: fundus at umbilicus → down a finger a day → not felt above pubis by ~2 weeks → near normal by 6; after-pains with feeding. **Lochia:** red → pink-brown → yellow-white. Perineum stretched or torn, episiotomy heals 1–2 weeks, early leakage; lax abdomen, diastasis; lax ligaments for months; **DVT risk raised 6 weeks**; milk comes in day 2–4; constipation; baby blues.

Goals: prevent complications (chest, DVT, retention, constipation); relieve pain; **restore pelvic floor and abdominal wall**; posture for feeding, lifting, carrying; emotional support; gradual return to fitness and work.

Postnatal programme (S16, W20)

- **Week 1:** deep breathing; ankle pumps; **pelvic floor from day 1** (eases swelling); transversus, pelvic tilts; roll to get up; walking; check diastasis; feeding posture.
- **Weeks 2–6:** longer holds, fast contractions; bridging, side-lying leg lifts, four-point arm and leg lifts; head lifts only if no doming; daily walks.
- **6–12 weeks:** after the 6-week check: brisk walking, cycling, swimming (bleeding stopped), resistance.
- **After 12 weeks:** running only when load and impact tests pass without leak, heaviness or pain (Goom 2019); 150 min/week.

Perineum: wrapped ice **10–15 min** in first 24–48 h; pelvic floor pumping; side-lying; soft cushion, squeeze before sitting; pad on stitches when opening bowels; no straining; report pain, smell, discharge.

Caesarean complications (3m S17)

Bleeding; wound or uterine infection; **DVT, PE**; chest infection, atelectasis; retention, bladder injury; **ileus, wind pain**; wound breakdown, hernia, adhesions; spinal headache; delayed mobility and feeding; later scar rupture.

After caesarean (S17 elective, cephalopelvic disproportion)

Before: breathing, supported huff and cough, bed mobility, leg exercises. **Day 0–1:** deep breathing; **huff with a pillow over the wound**; ankle pumps, static quads; roll and push up; early upright walking; gentle transversus, pelvic floor. **Days 2–7:** walking; feeding off the wound (**side-lying, football hold**, lap pillow); walking, pelvic tilts, abdominal massage for wind. **Weeks 2–6:** progress abdominal and pelvic floor; scar massage once healed; **lift nothing heavier than the baby ~6 weeks**; drive when an emergency stop is painless. **Home:** pelvic floor, transversus 3 times a day; warning signs: calf pain, fever, discharge, breathlessness.

Postnatal complications (LAQ S23, 7 + 8)

Perineal pain: ice, pelvic floor, positioning. **Retention, leakage:** walking, privacy; pelvic floor training. **Constipation, haemorrhoids, anal leakage:** fibre, footstool, no straining; sphincter training. **Diastasis:** transversus, no sit-ups while doming. **Back, pelvic girdle, coccyx pain:** posture, core, belt, cushion. **DVT:** walking, leg exercise; calf pain = urgent review. **Chest:** breathing, huffing. **Mastitis:** fever, red hot area → refer. **De Quervain's, carpal tunnel:** neutral wrist, splint. **Blues, depression:** screen, refer.

Back pain after delivery (W18; W21)

Differential: mechanical (feeding, lifting, weak core); pelvic girdle (sacroiliac, symphysis); disc with root pain; coccydynia; **dragging pain from pelvic floor weakness or prolapse** (W21 forceps: PERFECT, heaviness, pelvic floor training); rarely epidural-site infection. **Red flags:** fever, night pain, leg weakness, bladder or bowel change, **saddle numbness** → refer. **Assess:** delivery (instruments, epidural, baby weight), habits, posture, **active straight leg raise, posterior pelvic pain provocation**, neuro screen, diastasis, pelvic floor, mood.

Engorgement (3m S24), blues (3m S22), feeding ergonomics (3m S21)

Engorgement: painful hard breasts **2–5 days** after birth (milk stasis + swelling). Feed on demand **8–12 times a day**, correct latch; hand-express to soften areola; **cold packs** between feeds; gentle lymphatic strokes to the armpit, **no deep massage**; supportive bra; fever, red hot area → refer (breastfeeding protocol 2022).

Blues: tearful, mood swings, poor sleep in about half or more; starts **day 2–3**, settles **within 2 weeks** (hormone fall, sleep loss). Reassure, rest, family help. **Beyond 2 weeks or severe:** postnatal depression → **Edinburgh scale**, refer; confusion or hallucinations = psychosis, emergency.

Feeding: baby to breast, pillow on lap; back-supported chair, feet on stool, arm supported; look up between sucks; vary holds (cradle, cross-cradle, **football** after caesarean, side-lying at night); straight wrists; nappies at waist height; carrier high and close.

ICF (S26: 33, first baby, 15 days after normal delivery)

- **Structure/function:** dull back pain 5/10; heavy tender breasts; weak transversus, pelvic floor; possible diastasis; forward head; fatigue.
- **Activity:** long feeds, lifting, bathing, rising from floor or low bed. **Participation:** baby care, self-care, rest.
- **Environment:** family, ASHA lactation help (facilitators); floor sitting, low bed, no feeding pillow (barriers). **Personal:** anxious, little feeding knowledge.

Goals: short-term (1–2 weeks): engorgement settled, correct latch; pain **5 → 2/10**; correct pelvic floor and transversus; daily walks. Long-term (6–12 weeks): pain-free baby care; good pelvic floor; functional gap; **150 min/week**. **Doses:** heat **10–15 min**; transversus **10 × 10 s**, bridging **10**, cat–camel, pelvic tilts **2–3 times a day**; pelvic floor **≥ 8 contractions 3 times a day**; lift close with bent knees.

Exam plan (postnatal long case 15)

1. **ICF (5):** delivery details, impairments (pelvic floor, abdomen, back, breasts, perineum), activity, participation, factors; one-line diagnosis.
2. **Goals (3):** short and long term, measurable.
3. **Management (7):** early care, pelvic floor and abdominal doses, feeding ergonomics, complications, home programme, return to fitness.

Draw: stages of labour with PT; TENS sites T10–L1, S2–S4; mother feeding with lap pillow and footstool. **Underline:** upright, her choice, pant-pant-blow, involution, pelvic floor from day 1, supported huffing, baby to breast, red flags.

Menopause and osteoporosis

8 papers: changes at menopause, osteoporosis.

Full notes p. 63

Asked: 8 of 20 papers. **Long answers:** 45-year-old with menopausal symptoms, leakage, rib and back pain (W18); hormonal changes and sequelae with PT (W24, 8 + 7); 75-year-old with osteoporosis and knee pain (S25, 5 + 10). **7 marks:** importance of PT after menopause (S17, S22); physiological variations (S23). **3 marks:** prevention of osteoporosis (W18), predisposing factors (W22), types (S23), risk factors (S26).

Menopause and hormones (W24; 7m S23)

Permanent end of menstruation from loss of follicle activity, diagnosed after **12 months without a period**; India **~46 years** (West ~51). Premature: before 40; surgical: both ovaries removed (sudden). **Oestrogen and progesterone fall**; inhibin falls → **FSH rises markedly**, LH rises; androgens relatively higher.

Effects of low oestrogen

Vasomotor: hot flushes, night sweats, palpitations, poor sleep. **Mind:** mood swings, anxiety. **Bone: rapid loss** in early years → wrist, spine, hip fractures. **Muscle, joints:** less mass and strength, aches. **Genitourinary:** dryness, urgency, infections, **stress incontinence, prolapse**. **Heart, metabolism:** worse cholesterol, abdominal fat, BP. **Skin:** thin, weight gain. Medical: hormone therapy, bone drugs.

Importance of PT after menopause (7m S17, S22)

Bone: progressive resistance + impact, back extensors. **Falls:** balance, home safety, footwear. **Pelvic floor:** pelvic floor training, bladder training. **Weight, heart:** aerobic **150 min/week** + resistance. **Pain:** exercise, posture. **Flushes, sleep, mood:** exercise, relaxation, paced breathing, yoga; cool room. **Education:** calcium, vitamin D, sunlight, no smoking or alcohol, screening.

Osteoporosis: definition and types (3m S23)

WHO 1994: systemic skeletal disease with **low bone mass and microarchitectural deterioration** → fragility and fracture. **DEXA** hip and spine, T-score: normal **–1 or above**; osteopenia **–1 to –2.5**; osteoporosis **–2.5 or below**; + fragility fracture = established.

Type I (postmenopausal)	Women 50–70; trabecular; **vertebral crush, Colles'**
Type II (senile)	Both sexes over 70; trabecular + cortical; **hip**, vertebra
Others	Idiopathic juvenile; secondary (steroids, thyroid, parathyroid, rheumatoid arthritis, alcohol, immobility); regional (disuse, complex regional pain)

Risk factors (3m W22, S26), prevention (3m W18)

Fixed: female, age, early menopause, family hip fracture, thin build, previous fragility fracture. **Changeable:** inactivity, low weight, low calcium and vitamin D, smoking, alcohol, caffeine. **Disease, drugs:** steroids, rheumatoid arthritis, thyroid, diabetes. **Prevent:** peak bone mass in youth (jumping); lifelong impact, resistance, balance; calcium (milk, **ragi**, sesame, greens), sunlight; no smoking; falls prevention; DEXA, drugs if high risk.

Exercise for osteoporosis (S25)

Multicomponent; aerobic alone is not enough (Too Fit to Fracture 2014). **Resistance 2–3 days, 8–12 reps**, progress: sit-to-stand, step-ups, rows, wall push-ups, back extensors. **Impact** (no fracture): heel drops, stamping, stairs. **Balance most days:** tandem, single-leg, tai chi. **Posture:** chin tucks, thoracic extension. **Spine-sparing:** no loaded flexion or twist (toe touches, sit-ups); **hip hinge**; log-roll out of bed. **S25:** TUG, chair stand; higher chair, low step, rail.

ICF (W18: 45, 56 kg, 5'6", flushes, leakage, rib and back pain)

- **Structure/function:** perimenopause; back and rib pain (suspect osteoporotic fracture: **X-ray, DEXA**); leakage; hot flushes; BMI ~20.
- **Activity:** getting out of bed, lifting, sleep. **Participation:** work, family, social life.
- **Environment:** home, family support. **Personal:** 45, thin build.

Assess: red flags (sudden pain, height loss); posture; gentle rib springing; single-leg stance, TUG; pelvic floor (PERFECT); ICIQ leakage score. **Manage:** heat, TENS, log-roll; resistance **2–3 times a week**, daily walking, impact only after fracture excluded; posture **10 × 2 daily**; **pelvic floor ≥ 8 contractions 3 times a day**, the knack; aerobic, relaxation; diet, sunlight, balance.

Exam plan

7m importance of PT: changes (2) → PT per problem (4) → screen bone, spine-sparing (1). **Draw:** T-score scale; hip hinge vs rounded back. **Underline:** 12 months, FSH rises, T-score –2.5, resistance and balance, spine-sparing.

Revised ☐☐☐

Urogenital dysfunction: incontinence, prolapse and the pelvic floor

16 papers: incontinence and prolapse cases, PERFECT, Kegel's.

Full notes p. 70

Asked: long cases: 38-year-old leaking on coughing and lifting (W22, S25); 58-year-old, grade I prolapse, dribbling (W16); micturition + assessment + objective management (W19); bowel problems in pregnancy and after birth (S18); PCOD, BMI 34, back pain (S26). **7 marks:** types of stress incontinence and PT (S19); prolapse PT, grades, types and degrees, effects of exercise (S20, S21, W22, W25). **3 marks:** pelvic floor muscles (S16), continence advice (S18), IFT (W18), deferment (W19), PERFECT (W20), prolapse types (W21), hysterectomy indications (S23), Kegel's (S24), tests (S25).

Pelvic floor muscles (3m S16)

Pelvic diaphragm (deep)	**Levator ani:** pubococcygeus (with puborectalis, loops behind rectum), iliococcygeus; coccygeus
Deep perineal pouch	Deep transverse perineal, external urethral sphincter
Superficial pouch	Bulbospongiosus, ischiocavernosus, superficial transverse perineal
Others	External anal sphincter; **perineal body** = central tendon

Nerve: pudendal (S2–S4) + direct S3–S4. **Functions:** support; continence; relax to void; sexual; controls abdominal pressure with diaphragm and transversus.

Micturition (W19, 5 of 15)

Sympathetic (hypogastric, **T10–L2**): relaxes detrusor, closes bladder neck (storage). **Parasympathetic** (pelvic, **S2–S4**): contracts detrusor (emptying). **Pudendal** (**S2–S4**): external sphincter, voluntary holding. Capacity **400–600 ml**; cortex inhibits the **pontine micturition centre** until convenient → sphincters relax, detrusor contracts.

Incontinence types (7m S19)

Stress (SUI)	Leaks on effort, cough, sneeze: weak floor (childbirth, menopause, cough, obesity, lifting)
Urgency (UUI)	With or just after urgency: overactive detrusor, caffeine, infection
Mixed; others	Treat the worse first; overflow, fistula, functional

Stamey grades: 1 cough, sneeze, lift; 2 walking, standing up, stairs; 3 regardless of activity. **Mechanism:** urethral hypermobility or sphincter deficiency. **PT:** PFMT for all; knack for grade 1; biofeedback or ES if very weak; grade 3 → urogynaecologist.

Assessment and tests (W19, 3m S25)

History: when it leaks, how much, pads; urgency; bowel; births (big baby, forceps, tears), menopause, cough, weight, diuretics. **Refer:** blood in urine, recurrent infection, pain, palpable bladder, neuro signs, large prolapse.

- **Bladder diary** ≥ 3 days; **ICIQ-UI** short form.
- **Pad test:** 1-hour gain **≥ 1 g** = positive. **Cough stress test** with a comfortably full bladder.
- **Q-tip test:** swings **> 30°** on straining = hypermobility.
- Digital vaginal exam (consent): PERFECT; perineometer. Urine test, residual, urodynamics before surgery.

PERFECT (3m W20; Laycock and Jerwood 2001)

P-E-R-F-ECT

Power: Modified Oxford **0–5** (0 none, 1 flicker, 2 weak, 3 moderate with lift, 4 good, 5 strong). **E**ndurance: seconds held (up to 10). **R**epetitions of that hold (up to 10). **F**ast: 1-second contractions (up to 10). **ECT**: every contraction timed → becomes her programme.

PFMT: Kegel's (3m S24)

- **Supervised ≥ 3 months** = first-line for stress and mixed (NICE); ~**8 times** more likely cured (Dumoulin 2018).
- "**Squeeze and lift** as if stopping wind and urine"; no breath-holding or buttock squeeze; check by palpation; stopping the stream only to find the muscle.
- **Dose:** holds up to **10 s × 10** + up to **10 fast**; at least **8 contractions 3 times a day**, set from PERFECT.
- Lying → sitting → standing → functional (lift, stairs, jump). **The knack:** strong contraction just before and during cough, sneeze, lift.

Bladder training, deferment (3m W19), adjuncts (IFT 3m W18)

Bladder training ≥ 6 weeks: void by the clock, lengthen gap by ~**15 min** towards **3–4 h**. **Deferment:** stop, stand still or sit (perineal pressure); quick strong squeezes; slow breathing; distract; curl toes; walk calmly to the toilet.

Biofeedback	Perineometer or EMG if she cannot feel it
Electrical stim.	~**50 Hz** weak muscle (SUI); ~**10 Hz** calms overactive bladder; always with training
IFT (W18)	Two medium-frequency currents cross → low-frequency **beat** deep, no vaginal probe. Electrodes: under ischial tuberosities + anterior perineum (Laycock and Green), or 4 crosswise (2 suprapubic, 2 by tuberosities). Trials: carrier ~**4000 Hz**, beat **0–100 Hz**, **15 min, 3×/week, 15 sessions**. Weak evidence; NICE: only if she cannot contract, with PFMT. Avoid: pregnancy, infection, bleeding, cancer, pacemaker
Vaginal cones	Held while walking; muscle must contract

Continence-promoting advice (3m S18)

Drink normally (**1.5–2 L**); less caffeine, fizzy, alcohol; avoid constipation (fibre, footstool); weight loss if **BMI over 30**; stop smoking, treat cough; sit fully; knack before lifting; PFMT for life.

Prolapse (7m S20, S21, W22, W25; 3m W21)

Definition: descent of a vaginal wall, uterus or vault. **Anterior:** cystocele, urethrocele. **Middle:** uterine, vault. **Posterior:** rectocele, enterocele.

1st degree	Cervix down, inside vagina (POP-Q I: > 1 cm above hymen)
2nd degree	Cervix out through opening on straining (II: within 1 cm of hymen)
3rd (procidentia)	Whole uterus outside (III: > 1 cm below; IV: complete eversion)

Causes: childbirth, low oestrogen, chronic pressure (cough, constipation, lifting, obesity), ageing. **Symptoms:** something coming down, dragging worse by evening, backache, leaks.

PT for prolapse; effects of exercise (W25)

- Supervised PFMT **≥ 16 weeks** first option for stage 1–2 (NICE).
- **How it works:** knack narrows the **levator hiatus** during cough or lift; strength training → thicker, stiffer muscle, bladder and rectum **higher at rest** (ultrasound, Brækken 2010).
- **Effects:** one POP-Q stage better in **19% vs 8%** (Brækken); fewer symptoms (POPPY); Indian trial: one-to-one training beat a manual (Kashyap 2013). Reviews: first-line for stage I–III; long-term effect uncertain (Bø 2022). Dose: **3 × 8–12** near-max, **3–4 times a week**, ≥ 4 months. Modify, not stop, straining tasks.
- Weight, constipation, cough, knack; **pessary** if unfit or unwilling for surgery; surgery (vaginal hysterectomy, colporrhaphy) with PT before and after.

Bowel in pregnancy and after birth (S18)

Reasons: constipation (progesterone, iron, uterus pressure); haemorrhoids; painful defecation (stitches); anal incontinence (3rd–4th degree tear, pudendal stretch). **Assess:** diary, Bristol scale, leakage, tear. **Manage:** footstool, knees above hips, lean forward, breathe out, no strain, pad on stitches; fibre, fluids; gentle PFMT and sphincter exercises early.

Hysterectomy (3m S23), PCOD (S26)

Indications: fibroids (commonest), uncontrolled bleeding, adenomyosis, endometriosis, prolapse, cancer, chronic PID pain, postpartum haemorrhage. **PT:** breathing, supported cough, leg exercises, early walking, gentle PFMT, no heavy lifting ~**6 weeks**. **PCOD:** 2 of 3 (anovulation, androgen signs, cystic ovaries); aerobic **150–300 min** a week + strength 2 days; 5–10% weight loss over 6 months; core, posture.

ICF (W22, S25: 38, two children, leaks on cough, housework, lifting)

- **Structure/function:** weak pelvic floor (Oxford 2, endurance 4 s, 5 fast), no pre-contraction.
- **Activity:** coughing, sneezing, lifting, housework, exercise.
- **Participation:** avoids outings, classes, social events; pads.
- **Environment:** can afford PT, family (facilitators); distant toilets, pad cost (barriers). **Personal:** 38, embarrassment, maybe overweight.

Goals: short-term (4–6 weeks): correct contraction; Oxford 2 → 3; 6-s holds; knack; leaks halved. Long-term (3–6 months): Oxford 4; 10-s holds × 10; 10 fast; dry on cough, lift, housework; no pads. Low-impact aerobic **150 min a week**; review every **4–6 weeks**; refer if no gain by 3 months.

Exam plan (15-mark case)

1. **ICF diagnosis (5):** type with reasoning (cough, lift → SUI), six boxes, one-line diagnosis.
2. **Goals (3):** measurable by PERFECT, diary, pad test.
3. **Management (7):** PFMT with dose, knack, functional progression, adjuncts, bladder training if urgency, lifestyle, review and referral.

W16 prolapse case: add prolapse, bowel, menopause; inspect on straining → PFMT, knack, pessary or local oestrogen referral.

Draw: pelvic floor sling with three openings; three degrees of prolapse; ICF model. **Underline:** pudendal S2–S4, bladder diary, pad test, PERFECT, supervised 3 months, the knack, urge deferment, POP-Q.

Senior citizens in India: policies, NGOs, living arrangements and communication

6 papers: policies, institution vs home, communication barriers.

Full notes p. 85

Asked: long cases: 65-year-old alone on a fourth floor after Girdlestone arthroplasty, institution or home (S18); 60-year-old after Austin Moore prosthesis, hospital or home (S20). **7 marks:** national policies for the elderly (S18); communication barriers (S20, W21, W24). **3 marks:** policies (W18); role of NGOs (W19).

Senior citizens in India

Senior citizen: aged **60 or over** (2007 Act). **104 million** (8.6%) in Census 2011 → ~**227 million (14.9%) by 2036**; more elderly than under-15s by ~2046. Mostly rural; **feminisation of ageing**; oldest old (80+) growing fastest; most without pension.

Social issues: joint family break-up, elders alone; financial dependence; multiple diseases, poor access; loneliness, lost role; **elder abuse**; unsafe housing (stairs, Indian toilets).

Policies and schemes (7m S18; 3m W18)

National Policy on Older Persons 1999	Financial and food security, health, shelter, protection from abuse
Maintenance and Welfare Act 2007	Children must maintain parents; tribunals (up to **₹10,000 a month**); old age homes; abandonment punishable
NPHCE 2010	Geriatric wards at district hospitals; **rehab and PT at CHCs**; home care via PHCs
PM-JAY 70+ (2024)	**₹5 lakh** a year cover for all aged 70+ (Vay Vandana card)
Rashtriya Vayoshri Yojana 2017	**Free aids** (sticks, walkers, hearing aids, spectacles, dentures) at camps
Atal Vayo Abhyuday Yojana	NGO grants: homes, day care, mobile units; **Elderline 14567**

Also: **IGNOAPS** pension below poverty line; Ayushman Arogya Mandirs (elderly and palliative care); tax and travel benefits.

NGOs (3m W19)

HelpAge India, Dignity Foundation, Agewell. **Roles:** mobile units and camps (eye, hearing, PT); old age homes, day care; helplines, abuse protection; livelihood and pension help; sponsoring destitute elders; advocacy; caregiver training; linking to schemes.

Communication barriers (7m S20, W21, W24)

Hearing	**Presbycusis**: face them, quiet room, slow, lower pitch, no shouting, aid working
Vision	Good light, no glare, glasses, large high-contrast print
Cognition	Short sentences, one instruction, repeat, demonstrate, pictures, family
Speech	Aphasia, dentures: yes/no questions, gestures, picture boards
Emotional; physical	Fear: trust, listen; pain, fatigue: short sessions
Cultural; professional	Local language, simple words; no jargon, hurry or **elderspeak**, address them directly, **teach-back**
Environment	Quiet, well-lit, sit at the same level

Home vs institution vs hospital (S18, S20, W17)

Home: familiar, independence, dignity, cheaper; but unsafe stairs and toilet, loneliness, no emergency help. **Institution:** supervision, safety, meals, medicines, peers; but lost independence and privacy, separation, cost, depression. **Hospital:** acute illness only; deconditioning, delirium. **Decide by:** wishes, stability, ADL, cognition, falls risk, home, carer, money. Aim: **home with support where safe**.

ICF (S18: 65, alone, 4th floor chawl, left Girdlestone)

- **Structure/function:** femoral head and neck removed: **short leg**, weak unstable hip, pain. **Activity:** walking, stairs, toileting.
- **Participation:** living alone, shopping. **Environment:** 4th floor, no lift, shared toilet, alone. **Personal:** 65, independent.

Long-term goals (3–6 months): walks indoors and out with stick or walker and **shoe raise**; stairs with rail; independent self-care; no falls. **Management:** strengthen hip abductors, extensors, quads; walker → stick in right hand; stairs up with the good leg, down with the operated; balance; raised toilet seat, grab bars, phone in reach. **Answer (4m):** short stay with family or care facility during rehab, then home once stairs and toilet are safe; the choice is hers. **S20:** home with trained helper and home PT unless unstable.

Exam plan

Communication (7): why it matters (1: safety, adherence, dignity) → barriers (4) → one remedy each (2). **Policies (7):** table above + PT role in NPHCE. **Draw:** ICF model; home vs institution table. **Underline:** 60 and over, feminisation, 2007 Act, NPHCE, Vayoshri, presbycusis, teach-back.

Theories of ageing

9 papers, mostly 7 marks.

Full notes p. 92

Asked: 7 marks in 7 papers: define ageing, types, genetic theories (W16); theories of ageing (S17, S18, W19, S24); enumerate and explain any three (W20); enumerate and describe one (S21). **3 marks:** enumerate theories (S22); three theories (W22).

Ageing and its types (7m W16)

Ageing: progressive, time-related decline in structure and function that **reduces the ability to adapt to stress** and raises the risk of disease and death. **Universal, progressive, intrinsic, harmful. Types:** chronological; biological (two 70-year-olds differ); psychological; social (retirement, widowhood); functional; **primary** (age itself) vs **secondary** (disease, inactivity). **Successful ageing** (Rowe and Kahn): low disease and disability, high function, active engagement.

Programmed (genetic) theories (W16)

Genetic clock	Genes switch on and off in sequence; species lifespan; progeria
Hayflick limit	Fetal fibroblasts stop after **~50 doublings** (40–60) (Hayflick and Moorhead 1961); senescent cells build up
Telomere	TTAGGG caps **shorten each division** (end-replication problem) → senescence; ~9.5 kb at birth; **telomerase** rebuilds (germ, cancer cells); Nobel 2009. Against: mice long telomeres, short life; cancer risk
Endocrine (Dilman)	Hypothalamic clock: ↓ growth hormone, sex hormones, melatonin; menopause
Immunological (Walford)	Thymus shrinks (**immunosenescence**): more pneumonia, shingles, cancer; "inflammaging"; poor vaccine response

Damage (stochastic) theories

Wear and tear (Weismann 1882)	Tissues wear out with use (cartilage in OA); but body repairs and use strengthens
Cross-linking (Bjorksten 1942)	Collagen cross-links, glycation → **stiff** skin, arteries, tendons, lens
DNA damage; error catastrophe (Orgel 1963)	Unrepaired mutations (cancer rises with age); protein errors (largely not supported)
Rate of living (Pearl 1928)	Faster metabolism, shorter life; birds an exception

Free radical theory in full (7m S21)

Harman 1956; mitochondrial version 1972. Radical: **unpaired electron**. ROS: **superoxide, hydrogen peroxide, hydroxyl**, from normal metabolism (electron leak in mitochondria). Damage: membrane **lipid peroxidation**, oxidised proteins, **DNA** mutations; damaged mitochondria leak more (**vicious circle**). Defences: **superoxide dismutase, catalase, glutathione peroxidase**; vitamins C, E. **Lipofuscin** builds up. **Against:** antioxidant pills do not prolong life (68 trials); naked mole-rat lives > 28 years despite high damage. **PT:** regular exercise raises own antioxidant enzymes (hormesis); pills blunt gains.

Psychosocial theories

Disengagement (Cumming and Henry 1961): elder and society withdraw by mutual agreement; criticised: not inevitable, harms health. **Activity** (Havighurst 1961): staying active, new roles → **successful ageing**; basis of exercise groups. **Continuity** (Atchley 1989): keep earlier habits (temple, walking group). **Erikson's 8th stage: ego integrity vs despair**.

What they mean for PT

No single theory explains ageing: today seen as interacting **hallmarks of ageing** (12 in 2023). Much decline is **secondary**. **Cross-linking:** daily stretching; diabetes control. **Wear and tear:** joint protection, weight control; disuse is worse than use. **Telomere:** activity, no smoking, weight and stress control. **Immune, endocrine:** exercise improves immunity, insulin sensitivity; vaccination. **Activity, continuity:** groups, walking clubs, roles. **Disengagement:** withdrawal is a risk, not normal.

Exam plan (7 marks)

Definition (1) → **tree (2)** → **three theories (3)**: telomere, free radical, activity, each idea + evidence → **PT meaning (1)**. **Describe one:** free radical in full + circle diagram. **3 marks:** one line each from three branches.

Draw: classification tree; telomere caps shortening; free radical circle. **Underline:** Hayflick limit, telomerase, free radicals, superoxide dismutase, lipofuscin, cross-linking, immunosenescence, activity, successful ageing.

Revised ☐☐☐

Physiology of ageing and the effects of exercise

14 papers: changes in each body system and how exercise helps.

Full notes p. 101

Asked: long answers: cardiorespiratory and metabolic changes + national policies + fun, function, frailty, failure (W20); physiological effects of ageing + exercise on the neuromuscular system (W21); musculoskeletal and neurological changes + exercise (W22, 9 + 7). **7 marks:** nervous system (S17, S26), with exercise (W25); respiratory (S22); sensorimotor (S23). **3 marks:** visual (S16); musculoskeletal (W16, S25); cardiorespiratory (S19); six cardiopulmonary changes (S20, W24); respiratory (S26). No patient case on this topic alone.

Ageing in general

Ageing: gradual, progressive **decline in reserve** and ability to adapt to stress, raising risk of disease and death. **Primary** = change in everyone; **secondary** = added by disease, inactivity, smoking, poor nutrition: much "ageing" is secondary, so **exercise helps**. At rest may look normal; loss shows under stress (illness, surgery, heat, exercise, a fall). People and systems age at different rates.

Level (Guccione)	What the person can do
Fun	Work, home and leisure without restriction
Function	Most tasks, done differently; leisure limited
Frailty	Basic ADL use most of the reserve; needs help outside or at home
Failure	Needs help with basic and instrumental ADL; may be bedbound

PT keeps people on the higher steps and helps them climb back after illness.

Musculoskeletal (3m W16, S25; W22)

Muscle	**Sarcopenia:** loss of mass, esp. **type II (fast) fibres**; fewer motor units; fat infiltration; strength and power fall faster than mass → slow sit-to-stand, poor trip recovery, falls
Bone	Loss of mass and quality, fast after menopause; thin cortex, fewer trabeculae → **osteoporosis**, hip, wrist, spine fractures, kyphosis
Cartilage	Less water and proteoglycan, thinner, stiffer → OA, pain, stiffness
Tendon, ligament, capsule	More collagen cross-links, less elastic, slow healing → ↓ ROM, tendon injury
Discs	Dehydrate, lose height → stiff spine, spondylosis, stenosis
Posture, gait	Forward head, **thoracic kyphosis**, flexed hips and knees; slower, shorter steps, wider base, more double support → falls, restricted breathing

Nervous system (7m S17, S26, W25)

- **Brain:** volume falls (frontal, hippocampus); fewer synapses and dendrites; white matter change; ↓ **dopamine** (slow movement), ↓ **acetylcholine** (memory).
- **Cognition: slower processing and reaction time**; weaker working memory, divided attention; knowledge, vocabulary, judgement kept.
- **Motor control:** slower, less smooth, poor fine coordination, slower anticipatory postural adjustments. **Sleep:** lighter, broken, early waking.
- **Peripheral:** fewer motor and sensory fibres; slower conduction; ankle jerks reduced or absent.
- **Autonomic:** weak baroreflex (**postural hypotension**); poor temperature control; urgency, nocturia.

Senses (visual 3m S16) and sensorimotor (7m S23)

Vision	**Presbyopia**; small pupil, yellow lens; poor dark adaptation, glare tolerance, contrast, depth; narrow field; cataract, glaucoma, macular degeneration → misjudges steps, night falls: lighting, contrast step edges, eye checks
Hearing	**Presbycusis**: high tones lost first; speech in noise → face them, speak slowly
Touch, proprioception	Fewer receptors; ↓ position and vibration sense, most in feet → poor balance on uneven ground, in dark
Vestibular	Hair cell and fibre loss → dizziness on turning
Taste, smell	Reduced → poor appetite

Sensorimotor summary: slower sensing (fewer receptors, fibres) + slower processing (central) + slower, weaker response (fewer motor units, sarcopenia) → **late, weak balance reactions** → falls (falls (p. 21)).

Cardiovascular and respiratory (3m S19, S20, W24, S26; 7m S22)

- **Heart, vessels:** stiff arteries → ↑ systolic BP, pulse pressure; thick LV, slow relaxation → filling depends on atrium (AF poorly tolerated); **max HR falls** (~220 – age); less adrenaline response; resting HR little change.
- **VO_2max** and max output fall: **3–6% per decade** in 20s–30s, **over 20% per decade** from the 70s (Fleg 2005). Varicose veins, swelling.
- **Chest wall** stiffer (cartilage calcification, kyphosis): rib cage compliance **~30% lower**. **Diaphragm** strength **~25% lower** at 65–75. **Lungs** lose elastic recoil; small airways close earlier.
- TLC same; **RV up ~50%** (20 to 70); VC falls; **FEV_1 –20–30 ml/year**; FEV_1/FVC falls (fixed 0.70 over-diagnoses COPD). Airways close in tidal breathing at **~44 lying, ~65 sitting**.
- Alveolar area **75 → 60 m^2** (30 to 70); V/Q mismatch: **$PaO_2 \approx 109 - 0.43 \times age$** (lying); $PaCO_2$ unchanged. Response to hypoxia **–50%**, hypercapnia –40–50%; less awareness of bronchospasm.
- Weaker, less sensitive cough; slow mucociliary clearance; **silent aspiration** (71% of older pneumonia cases vs 10%).
- **PT (7m S22):** lung complications after surgery 2–3 times more at 60–79 → pre-op breathing, **early mobilisation**; **sit up** (PaO_2 ~5 mmHg higher than lying); inspiratory muscle training (+26 cmH_2O); aerobic (VO_2max +16%); posture; airway clearance; monitor SpO_2 and Borg.

Six cardiopulmonary changes	Result
Stiffer arteries	Raised systolic BP
Lower max HR	Lower max output, VO_2max
Weaker baroreflex	Postural hypotension
Stiff chest, weak muscles	More work of breathing

Six cardiopulmonary changes	Result
Loss of lung recoil	↓ FEV_1, VC; ↑ RV
Weak cough, clearance	More chest infections

Metabolic and other (W20)

Less muscle and water, **more (abdominal) fat**; **BMR falls** (mostly from muscle loss); **insulin resistance**, type 2 diabetes; ↓ growth hormone, sex hormones, skin vitamin D; less sweating, shivering, thirst (heat stroke, cold, dehydration); ↓ kidney filtration, slow drug clearance; thin fragile skin (pressure sores; care with heat and electrotherapy); weaker immunity, poorer vaccine response.

Effects of exercise (W21, W22, 7m W25)

Muscle	Resistance training gains strength and power at any age: frail ~90-year-olds **+174% strength**, +9% thigh area, tandem gait +48% in **8 weeks** (Fiatarone 1990); early gains **neural**, later size
Nervous system	Faster reaction time, better coordination and balance reactions; aerobic: brain blood flow, memory, mood, less cognitive decline
Balance, falls	Exercise cut community falls rate by **23%**; balance + functional work best (Sherrington 2019)
Bone, joints	Weight-bearing and resistance slow bone loss; less OA pain, stiffness; stretching improves ROM
Heart, lungs	↑ VO_2max; ↓ resting HR and BP; better tolerance
Metabolism	Better insulin sensitivity, less abdominal fat, better sleep
Function, mind	Faster walking, easier stairs and sit-to-stand, independence; less depression, more social contact

WHO 2020 (65 and over)

- **150–300 min a week moderate** aerobic or **75–150 min vigorous**.
- Muscle-strengthening on **2 or more days** a week.
- **Multicomponent** balance and strength on **3 or more days** a week (function, falls).
- Some is better than none; reduce sitting.

Precautions

Screen heart disease, uncontrolled BP, diabetes, osteoporosis, falls risk; **beta-blockers blunt HR: use Borg**. Longer warm-up and cool-down; no breath-holding; rise slowly (postural hypotension); hydrate, avoid heat. Start low, progress slowly; supervise balance work; good footwear, safe floor.

Exam plan

W22 long answer: introduction (1: ageing, primary vs secondary, reserve) → musculoskeletal (4) → neurological (4) → exercise (7: resistance, balance, aerobic, flexibility; Fiatarone, Sherrington; WHO doses; precautions). **W20:** cardiorespiratory + metabolic + fun-to-failure + policies (elderly in India (p. 16)). **7 marks, one system:** 6–8 changes in a table (4) → effect on function (2) → PT implication (1).

Draw: fun-to-failure steps; young vs old posture (forward head, kyphosis, flexed knees). **Underline:** loss of reserve, primary and secondary, sarcopenia, type II fibres, slower reaction time, presbyopia, presbycusis, postural hypotension, lower max HR, ↑ residual volume, insulin resistance, exercise reverses.

Revised ☐☐☐

Geriatric evaluation and the role of physiotherapy

9 papers: assessment scales, functional assessment, geriatric rehabilitation.

Full notes p. 110

Asked: long cases: widow in an old age home with knee pain and falls (S17); Colles' wrist stiffness (W16); 69-year-old in a charitable centre (S19); geriatric rehab principles + policies (W19); neck pain (S21). **7 marks:** assessment scales (W17); rehab principles (W18); functional assessment (W18, W19); fitness assessment (S25). **3 marks:** immobility effects (S23).

Functional assessment: domains and scales (7m W17, W18, W19)

Comprehensive geriatric assessment: multidimensional team assessment of medical, functional, psychological and social needs. History: drugs (number, sedatives, BP), **falls in last year**, continence, nutrition, goals. Physical: **BP lying and standing**, vision, hearing, feet, grip, sensation.

Basic ADL	**Katz** 0–6 (6 independent, ≤ 2 very dependent); **Barthel** 0–100; FIM
IADL	**Lawton** 0–8: phone, shopping, food, housework, laundry, transport, medicines, money
Mobility, balance	**TUG ≥ 13.5 s** falls risk; **Berg** 0–56, below ~45; **Tinetti** 0–28, below 19; SPPB 0–12; gait speed below ~0.8 m/s
Cognition	**MMSE** 0–30, ≤ 23 impaired; Mini-Cog, MoCA
Mood	**GDS-15** > 5 depression; **FES-I** 16–64 fear of falling
Frailty, nutrition	**Fried:** weight loss, exhaustion, weak grip, slow walk, low activity; ≥ 3 frail, 1–2 pre-frail; MNA; Clinical Frailty Scale
Social, home	Carers, money, floor, stairs, toilet type, lighting: home visit

Fitness assessment (7m S25): Senior Fitness Test

30-s chair stand (leg strength; 43 cm seat, arms crossed); **30-s arm curl** (5 lb women, 8 lb men); **6-min walk** (50-yd course) or 2-min step; **chair sit-and-reach**; **back scratch**; **8-foot up-and-go** (agility); BMI. Add grip, single-leg stance, gait speed. **Normal range** = middle 50% (chair stand at 60–64: women 12–17, men 14–19). **2013 standards** to stay independent to ~90: women 60–64 **15 stands, 625 yd**; at 90–94 both 9 stands, 400 yd. Screen; stop for chest pain, dizziness, breathlessness.

Principles of geriatric rehab (7m W18, W19)

Comprehensive assessment; treat the person, not the age; **function is the goal**, set with them; teamwork; **mobilise early**; **start low, go slow, but progress**; prevent the next problem (falls, sores, contractures); environment and aids; comorbidity and drugs; simple instructions, carers; dignity and choice; continuity with community groups. **PT role:** groups, screening, treatment, home visits, carers.

Immobility in the elderly (3m S23)

Rapid loss of strength and walking (hospital-associated disability), contractures; postural hypotension, DVT, pneumonia; bone loss, pressure sores; incontinence, constipation, impaction; **delirium**, depression, fear of falling; insulin resistance, weight loss. Prevent: sit out and walk early, bed exercise, turning, toileting schedule.

ICF (S17: 65, widowed, old age home, left knee pain, 2 falls in a month)

- **Structure/function:** knee OA: pain, stiffness, weak quads and hip abductors, poor balance, low mood.
- **Activity:** stairs, squatting (Indian toilet), walking. **Participation:** life in the old age home.
- **Environment:** stairs to her room, Indian toilet. **Personal:** 65, widowed, wants quality of life.

Assess: falls history, postural BP, knee, TUG, Berg, chair stand, Barthel, MMSE, GDS, FES-I, home.

Pain	Heat or ice, TENS for flares; OA education
Weakness	Static quads, SLR, sit-to-stand, mini-squats, hip abduction; **10 × 2–3 sets, 3 times a week**, ankle weights
Balance	Narrow, tandem, single-leg at rail, stepping, turns; **over 3 h a week** in total
Stairs, toilet	Up with the good, down with the painful; raised commode or grab bar; stick in right hand
Fear, mood, fitness	Group exercise, getting up from the floor, grief counselling; walks to 30 min

Exam plan

Functional assessment (7): why (1) → domains with a scale each (4) → use: goals, falls and frailty risk, referral, review (2). **Draw:** ICF model; TUG course (chair, 3 m, turn). **Underline:** ADL and IADL, Katz, Lawton, Barthel, MMSE 23, GDS, TUG 13.5 s, Berg 45, Senior Fitness Test, start low go slow, function is the goal.

Revised ☐☐☐

Falls in the elderly

8 papers, with 5 long cases on fall risk and prevention.

Full notes p. 119

Asked: long answers: 88-year-old after hip replacement, home or old age home, prevention (W17); balance pathomechanics + prevention programme (S20, W24, 7 + 8); causes + rehab, 72-year-old with fear of falling (S23); intrinsic and extrinsic, two measures each (S24). **7 marks:** causes + environmental modifications (W22). **3 marks:** risk factors (W20); extrinsic factors (W25).

Definition and burden

Fall (WHO): event in which a person comes to rest **inadvertently** on the ground, floor or other lower level. **28–35%** of 65+ fall each year, **32–42%** over 70; a **previous fall** is the strongest predictor. Fractures, head injury, **long lie**, fear, lost independence, death.

Risk factors (3m W20, W25; W22, S23, S24)

Intrinsic	Leg weakness (quads, hip abductors, dorsiflexors), arthritis; poor balance, shuffling gait, previous falls; vision (cataract, bifocals), foot neuropathy, vestibular loss; **postural hypotension**, stroke, PD, dementia, depression, nocturia; fear
Extrinsic	Poor light, wet floors, loose rugs, clutter, cords, no stair rails, low beds, **Indian toilets**, no grab bars, uneven roads; **4 or more drugs**, sedatives, antidepressants, antipsychotics, BP drugs, diuretics; loose slippers, worn soles, faulty aids

Two measures each (S24): strength, footwear; balance, aid; eye check, single-vision glasses; medical review, rise slowly; home check; drug review.

Pathomechanics of balance (7m S20, W24)

- **Sensory:** vision (contrast, glare), vestibular (hair cells), feet less sensitive; more reliance on vision, so dim light falls.
- **Central:** slower reaction; **dual-tasking** worsens gait.
- **Motor:** type II fibre and power loss, step too slow or short; ↓ ankle range → **ankle to hip strategy**, late stepping.
- Flexed posture, smaller limits of stability; late anticipatory adjustments; postural hypotension, fear, drugs → **late, weak response: fall**.

Assessing risk

Three questions: fallen in the past year? unsteady? worried? Each fall: where, how, dizziness, injury, time on floor. **TUG ≥ 13.5 s**, Berg, gait speed, chair stand, postural BP, vision, feet, cognition, FES-I; drugs; home visit. Low risk: education + exercise; high risk (injury, repeated falls, cannot get up, poor gait): **multifactorial plan** (World Falls Guidelines 2022).

Prevention programme (8m S20, W24; W17)

- **Exercise** cut falls **23%**; balance and functional 24%; with resistance about a third (Sherrington 2019); best with **over 3 h a week** of challenging balance (2017).
- **Otago:** home leg strengthening (ankle weights) + balance **~3 times a week** + walking; falls down about a third.
- **Balance:** narrow base (tandem, single leg), less hand support, reaching, stepping, turns, tai chi; add dual tasks.
- **Strength:** sit-to-stand, mini-squats, heel and toe raises, hip abduction, knee extension; **2–3 sets of 8–12, 2–3 times a week**.
- Drug review; cataract surgery; footwear; rise in stages; vitamin D, hip protectors in care homes.

Environment (7m W22): bright light, night lights to toilet; no rugs or cords, mark step edges; rails both sides; grab bars, **raised commode over Indian toilet**, bathing stool; bed and chair at knee height; items waist to shoulder height. Home modification: rate ratio **0.81** (Gillespie 2012). **Getting up:** check injury → roll to side → hands and knees → crawl to chair → half-kneel, push up.

ICF (S23: 72, male, two falls in a year, fear of falling)

- **Structure/function:** poor balance, weak legs, fear. **Activity:** walking, stairs, going out.
- **Participation:** social life, independence. **Environment:** home hazards, drugs. **Personal:** 72, lost confidence; check BP drugs, nocturia.

Rehab: balance 3 times a week supervised + daily home, to **over 3 h a week**; Otago strength 2–3 × 8–12; gait with turns, obstacles, dual tasks; **fear:** small goals, graded exposure, floor recovery, groups; home and drug review; alarm in reach. Review **every 4–6 weeks** (TUG, Berg, FES-I); continue long term.

Exam plan (8 marks: programme)
Screening and assessment (2) → exercise with dose and evidence (3) → home, drugs, vision, feet, BP, bone (2) → education, getting up, alarms (1). **Draw:** balance system; safe bathroom; getting-up steps. **Underline:** intrinsic and extrinsic, previous falls, psychotropic drugs, ankle to hip strategy, dual-tasking, over 3 hours a week, Otago, fear of falling.

Revised ☐☐☐

Disability: definitions, models, types, evaluation and prevention

9 papers: impairment, disability and handicap; levels of prevention.

Full notes p. 127

Asked: 7 marks: levels of disability prevention (W20, S26); types of disability + upper-limb permanent impairment (S23). **3 marks:** impairment, disability, handicap (W16, W25); levels of prevention (W16, W22, W25); ICIDH and models (W17); types (S19, W20); primary and secondary disability (S16).

ICIDH terms (3m W16, W17, W20, W25)

ICIDH: International Classification of Impairments, Disabilities and Handicaps, **WHO 1980**; replaced by the **ICF 2001**. Chain: disease → impairment → disability → handicap.

- **Impairment (organ):** any loss or abnormality of psychological, physiological or anatomical structure or function.
- **Disability (person):** any restriction or lack, from an impairment, of ability to perform an activity in the manner or range considered normal.
- **Handicap (society):** a disadvantage, from an impairment or disability, that limits or prevents a **role** normal for that individual (age, sex, social, cultural factors).

Polio → paralysed leg → cannot walk far → cannot farm. **ICF:** two-way, adds environmental and personal factors; disability = **umbrella term** for impairments, activity limitations, participation restrictions.

Models (3m W17) and types (3m S19, W20)

Medical: problem of the person → treat the impairment. **Social:** created by **barriers** (steps, attitudes, laws) → remove them. **Biopsychosocial (ICF):** interaction → treat person, change environment. **Rights-based:** UN Convention 2006, RPWD Act 2016. **Charity:** pity, dependence. **Nagi:** pathology → impairment → functional limitation → disability.

RPWD Act 2016: 21 specified disabilities. Locomotor (locomotor, leprosy-cured, CP, dwarfism, muscular dystrophy, acid attack); sensory (blindness, low vision, deaf and hard of hearing, speech and language); intellectual (intellectual disability, specific learning disabilities, autism); mental illness; chronic neurological (MS, Parkinson's); blood (haemophilia, thalassaemia, sickle cell); multiple incl. deaf-blindness. **Benchmark: 40% or more**, certified; needed for benefits. Also congenital or acquired, temporary or permanent, visible or invisible, mild to profound.

Primary and secondary disability (3m S16)

Primary: caused directly by the disease or injury (paraplegia after SCI). **Secondary:** later, from the primary disability, inactivity or poor care: contractures, pressure sores, obesity, osteoporosis, pain, UTI, depression, deconditioning; **largely preventable** by PT (positioning, exercise, education, early mobility).

Upper-limb permanent impairment (7m S23)

At **maximum improvement**; measured, not opinion. **Arm 90%**: ROM (shoulder, elbow, wrist **30% each**, mean % loss × 0.3); strength (MRC → % loss × 0.3); coordination (10 activities, **9% each**). **Hand 90%**: prehension 30% (opposition 8%, lateral pinch 5%, cylindrical 6%, spherical 6%, hook 5%); sensation 30% (thumb ray 9%, index 6%, others 5%); strength 30% (grip 20%, pinch 10%).

Combining formula: a + b(90 − a) ÷ 90 (a = higher). Arm values combined, hand values added, then arm and hand combined: arm 27%, hand 64% → **71.8%**. Extra up to **10%** (pain, deformity, dominant limb 4%, shortening 2% per inch beyond the first); never over 100%.

Levels of prevention (7m W20, S26; 3m W16, W22, W25)

WHO 1981; each level breaks one link of the chain.

Primary: no impairment	Immunisation (polio, measles, rubella); iodine, vitamin A, folic acid, iron; antenatal care; genetic counselling; helmets, seat belts; health education
Secondary: impairment not → disability	Screening (RBSK, hearing, vision, diabetic foot); early treatment; **early PT, splinting**; early surgery (clubfoot, cataract)
Tertiary: disability not → handicap	Rehabilitation; aids; prevent secondary conditions; inclusive education; access; jobs; rights; CBR (p. 26)

PT: education and exercise; early detection and therapy; rehab and inclusion. **Polio:** vaccine → early PT, calliper → rehab, school access.

Exam plan (prevention, 7 marks)

Chain (2) with terms → **three levels (4):** aim + five measures → **PT role (1)**. **Draw:** ICIDH chain with levels; ICF two-way. **Underline:** organ, person, society, umbrella term, social model, 21, benchmark 40%, secondary conditions, combining formula.

Revised ☐☐☐

Rehabilitation: types, IBR vs CBR and the rehabilitation team

15 papers: IBR vs CBR, NGOs, social worker, vocational rehabilitation.

Full notes p. 133

Asked: Long answer (S23, 2 + 3 + 10): define rehabilitation, team effort in CBR, CBR vs IBR. **7 marks:** IBR vs CBR (W16, W21, W24, S24, S25); vocational guidance (W17, S23); ADL and disabled hand (W17); social worker (S21); NGOs (S19). **3 marks:** rehabilitation (S20); social worker (W20, S26); OT (S26); vocational rehabilitation (W19, S21); NGOs (S16, W17, S23, S25); Spastics Society (S18); Red Cross (S20, W21). No patient case.

Definition and types (3m S20; S23)

Rehabilitation (WHO): "a set of interventions designed to **optimize functioning and reduce disability** in individuals with health conditions in interaction with their environment". **Older WHO wording:** all measures that reduce the impact of disabling and handicapping conditions and enable disabled people to achieve **social integration**.

Aim: highest possible independence. **Kinds:** medical, vocational, social, educational, psychological.

By place: IBR: hospitals, centres, special schools. **Outreach**: staff go out (camps, mobile vans, home visits). **CBR**: in the person's own community with family and local workers, **local resources**, referral when needed.

IBR vs CBR (7m W16, W21, W24, S24, S25; 10m S23)

CBR (ILO, UNESCO, WHO 2004): "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", through the **combined efforts** of disabled people, families, communities and government and non-government services.

	IBR	CBR
Place	Hospital, centre, special school; cities	Home, village, school, workplace
Who	Specialists	CBR workers, family, ASHA, anganwadi, teachers, volunteers + professionals
Approach	Medical model: impairment	Social and rights model: health, education, livelihood, social, empowerment
Decisions	Professionals; person a patient	With person and family as **partners**
Cost	High (buildings, staff, travel, stay)	Low; local materials, existing services
Coverage	Few, mainly urban	Many, incl. rural and poor
Technology	Advanced equipment, surgery	Simple, appropriate (bamboo or wooden aids, home adaptations)
Carry-over	Artificial setting	Real home and village
Continuity	Ends at discharge	Ongoing, community-sustained
Inclusion	May separate from family	Keeps person in family; changes attitudes
Limits	Expensive, crowded, far, dependency	Limited specialist skill; needs training, community commitment

Best for: IBR acute and complex care, prostheses, staff training; CBR long-term needs, prevention, inclusion. **Conclusion:** not rivals; CBR **refers** complex cases, institutions train and back up CBR workers, **outreach** links them (CBR (p. 26)).

Rehabilitation team (S23, S26)

Person and family at the centre, setting goals. **Physiatrist:** diagnosis, medical care, prognosis, certification. **Physiotherapist:** assessment, exercise, gait, pain, chest care, aids, home programme, prevention. **Speech therapist, audiologist:** speech, swallowing, hearing aids. **Prosthetist and orthotist:** limbs, callipers, splints. **Psychologist:** IQ tests, counselling. **Vocational counsellor:** work ability, placement. **Nurse:** skin, bladder, bowel, teaching family. Plus special teacher; in CBR, the CBR worker.

Multidisciplinary: each works separately. **Interdisciplinary:** share findings, common goals. **Transdisciplinary:** one **key worker** (CBR worker or family) delivers, trained by the others; usual in CBR.

Team effort in CBR (3m S23): needs span health, education, work, social life; shared goals avoid conflicting advice; few professionals train many local workers; two-way referral, continuity; community ownership → cheap, sustainable; family empowered.

Occupational therapist (3m S26) and social worker (3m W20, S26; 7m S21)

- **OT:** ADL assessment and training (feeding, dressing, bathing, toileting, cooking); hand function, splints, sensory re-education; adaptive devices (built-up spoon, button hook, reacher, sock aid); home, school, work modification; cognitive and perceptual retraining; pre-vocational skills, seating; sensory integration in children.
- **Medical social worker:** psychosocial assessment (family, income, housing, work); counselling to accept disability; **linking to benefits**: certificate and UDID card, pensions, concessions, free aids (ADIP), scholarships, loans; discharge planning, return to school or work; advocacy.
- **In CBR:** survey to find persons with disability; awareness; self-help groups, CBR committees; linking to schemes, livelihood, education.

Vocational rehabilitation (3m W19, S21; 7m W17, S23)

Purpose (ILO Convention 159, 1983): "to enable a disabled person to **secure, retain and advance in suitable employment**" and so further integration into society.

1. **Evaluation:** physical capacity (FCE), skills, interests, aptitude, work history, job market.
2. **Guidance and counselling:** realistic goal matching ability and interest.
3. **Training:** ITIs, Vocational Rehabilitation Centres for Handicapped, NGOs; work conditioning, work hardening.
4. **Placement:** old job with changes; open employment (**4%** of government posts, RPWD Act 2016); sheltered workshop; supported employment; self-employment with loans (National Divyangjan Finance and Development Corporation).

5. **Follow-up** to keep the job.

Significance of guidance: matches job to remaining ability (no wasted training); motivation, acceptance; earning, independence; social role, dignity; guides family and employer on adaptations; informs about schemes, reservation, loans.

ADL training and the disabled hand (7m W17 each)

Basic ADL: feeding, grooming, bathing, dressing, toileting, transfers, mobility. **Instrumental:** cooking, shopping, money, phone, transport. **Steps:** assess (**Barthel, FIM**) → goals with the person → task analysis → teach step by step (**forward or backward chaining**) in the real setting at the usual time → easiest method first, fade help → aids and environment → train family → review. Energy conservation and joint protection.

Examples: hemiplegia: affected arm in first, out last; paraplegia: sliding-board transfer, long-handled mirror; RA: built-up handles, lever taps; low vision: tactile marks on the stove.

Disabled hand (restore, protect, adapt): *assess* cause (nerve injury, leprosy, burns, stroke, RA), deformity, oedema (volumeter), goniometry, grip dynamometer (sitting, elbow 90°, mean of 3) and pinch gauge, monofilaments, two-point discrimination, Nine-Hole Peg, Jebsen–Taylor, DASH. *Grips* (Napier): power (cylindrical, spherical, hook), precision (tip, three-jaw, key pinch); train with graded objects. *Treat* oedema, mobilise, strengthen; splints (static or dynamic; bamboo or tubing): **cock-up** radial palsy, **knuckle-bender** mobile claw, web spacer median palsy, **safe position** for burns (wrist 20–30° ext, MCP 70–90° flex, IP straight). *Sensory re-education:* localisation then textures and objects, eyes closed, check eyes open. *Anaesthetic hand* (NLEP): inspect daily, soak ~½ h, scrape, oil while wet, protect from heat and friction. *Adapt*: built-up handles, universal cuff, one-handed use. *Work:* job analysis, work simulation, insulated tools, job change; tendon transfer for mobile claw.

NGOs (3m S16, W17, S23, S25; 7m S19), SSI (S18), Red Cross (S20, W21)

NGO: voluntary, non-profit, independent of government, a registered society or trust: ADAPT, Indian Red Cross, HelpAge India, Jaipur foot (BMVSS), Mobility India. **Role:** surveys, early identification; awareness; services where government does not reach (therapy, camps, aids, schools); training CBR workers and parents; vocational training, self-help groups; advocacy, linking to schemes; pilot programmes.

Spastics Society of India (now ADAPT): 2 October 1972, Mumbai, **Dr Mithu Alur**, for children with **CP**; Able Disabled All People Together. Inclusive education; therapies; teacher training; vocational training; early intervention; advocacy; outreach.

Red Cross: Henry Dunant (Solferino); ICRC Geneva **1863**; Indian Red Cross by an **Act of 1920**, President of India is president. **Seven principles:** humanity, impartiality, neutrality, independence, voluntary service, unity, universality. **Functions:** disaster relief; blood banks; first aid training; ambulances; maternal and child welfare; war victims; Junior Red Cross.

Exam plan

IBR vs CBR (7): definitions of rehabilitation and CBR (1) → table, at least 8 rows: place, who, approach, decisions, cost, coverage, technology, continuity, inclusion (5) → complement through referral and outreach (1).

S23 (2 + 3 + 10): WHO definition + kinds → team diagram, transdisciplinary, reasons → CBR definition, 10-row table, conclusion. Any disability long case: name the team and vocational steps.

Draw: team around person and family; IBR, outreach, CBR as three circles. **Underline:** optimize functioning, local resources, coverage and cost, social inclusion, referral, transdisciplinary, placement, advocacy.

Revised ☐☐☐

National policies and legal rights for persons with disability

7 papers: PWD Act, national programmes, barriers, concessions.

Full notes p. 144

Asked: Long case (S18, 5 + 5 + 5): 35-year-old farmer, T12–L1 injury: functional diagnosis, transport, concessions (W18 amputee case asks the same). **7 marks:** architectural barriers and modifications (S18). **3 marks:** PWD Act (S18, S23, S24); three national programmes preventing disability (S20, W21, W24); difficulties in the community (W17).

The Acts (3m S18, S23, S24)

	PWD Act 1995	RPWD Act 2016
Name	Equal Opportunities, Protection of Rights and Full Participation	Rights of Persons with Disabilities (UN Convention 2006)
Covers	**7**: blindness, low vision, leprosy-cured, hearing, locomotor, mental retardation, mental illness	**21** specified disabilities
Benchmark	40% or more	40% or more
Jobs	**3%**	**4%**
Education	Free to 18; 3% seats	Free 6–18; **5%** higher education; inclusive
Other	Prevention, early detection, barrier-free transport, aids, unemployment allowance, Commissioners	Equality, accessibility (buildings, transport, ICT), health, social security, special courts, penalties

Also: Rehabilitation Council of India Act 1992 (registers rehab professionals); National Trust Act 1999 (autism, CP, intellectual and multiple disability; Niramaya insurance); Mental Healthcare Act 2017; National Policy 2006.

National programmes (3m S20, W21, W24)

UIP (1985) + Pulse Polio (1995): polio, measles, rubella, diphtheria, tetanus. **NPCBVI (1976):** cataract, refractive error, glaucoma. **NPPCD (2006):** deafness: early detection, hearing aids. **NLEP (1983):** leprosy deformity, multidrug therapy. **Iodine deficiency programme:** iodised salt. **RBSK (2013):** screens defects, diseases, deficiencies, developmental delays; early intervention centres. **NPHCE (2010):** elderly. NCD and mental health (1982) programmes.

Difficulties in the community (3m W17)

Physical (steps, narrow doors, toilets, no ramps); **transport**; **attitudes** (stigma, pity, overprotection, marriage); **economic** (unemployment, cost of aids); **services** (distance, no information, paperwork); **education and communication**; **social** (isolation, dependence, abuse).

Architectural barriers (7m S18)

- **Ramp: 1:12** max, **1,200 mm** wide, handrails both sides (760–900 mm), landings top and bottom.
- **Doors ~900 mm**, lever handles, no high thresholds; **turning circle 1,500 mm**.
- **Toilet:** western seat at wheelchair height, grab bars, basin without pedestal, outward door, non-slip floor.
- Lift with low buttons or ground-floor rooms; lowered counters and switches; kerb cuts, tactile paving, audible signals; large contrast signs, Braille.

Harmonised Guidelines 2021; Accessible India Campaign (2015).

Long case (S18): farmer, T12–L1, discharged

Functional diagnosis: paraplegia (ASIA), strong arms and trunk; neurogenic bladder and bowel; pressure sore risk; independent wheelchair and transfers, short walks with orthoses; rural roads limit mobility; cannot farm.

Transport: wheelchair (pneumatic tyres, cushion, wheelie skills); **hand tricycle** (ADIP); motorised tricycle with hand controls, licence; car with hand controls; low-floor bus, train, auto transfers; knee–ankle–foot orthoses and crutches for short distances only.

Concessions: certificate and **UDID**; **ADIP** free aids; **rail 75%** (second class, sleeper, first, 3-tier AC, chair car), 50% first and second AC, escort same; state bus passes; disability pension (Indira Gandhi national scheme, BPL; Sanjay Gandhi Niradhar in Maharashtra); **80U** tax deduction (₹75,000; ₹1,25,000 severe); NDFDC loans (dairy, shop); 4% jobs; housing priority; panchayat ramp and toilet.

Exam plan (barriers, 7 marks)
Intro (1): barriers cause handicap (social model); RPWD accessibility duty → **barriers + modifications with measurements (5)** → **PT role (1):** home and building access audit. **Draw:** ramp 1:12 with handrails and landings; toilet with grab bars and turning circle. **Underline:** 7 vs 21, 3% vs 4%, 40%, UDID, ADIP, rail 75%, 1:12, 1,500 mm.

Revised ☐☐☐

Community-based rehabilitation: principles, strategies and the physiotherapist

11 papers: principles of CBR, the anganwadi worker, rural and urban strategies.

Full notes p. 150

Asked: Long answer (S22, 5 + 10): 6-year-old with CP found at a village screening, parents want school: local resources and CBR plan. **7 marks:** principles and team members (S17); rural and urban strategies (W19); teamwork benefits (S20); note on CBR (W20); PT in a school for physically handicapped children (W22); approaches and the anganwadi worker (S26). **3 marks:** aspects (S16); anganwadi worker (S17, S22); principles (W21, S22, S25); three features (W24).

Definition and features (3m S16, W24; 7m W20)

CBR (ILO, UNESCO, WHO 2004): "a strategy within general **community development** for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", carried out by disabled people, families and communities with health, education, vocational and social services.

History: WHO promoted CBR after **Alma-Ata (1978)**; joint papers 1994, 2004 widened it to rights and inclusion; **WHO CBR guidelines 2010** gave the matrix. Compare with IBR (p. 23).

- In the person's own village or town; **local resources**, simple technology.
- Person and family are **active partners**.
- Community workers (CBR workers, anganwadi workers, ASHAs, teachers) do most work, trained by professionals.
- Health, education, livelihood, social, empowerment, not only therapy.
- Works with PHCs, schools, panchayats; refers; low cost, wide coverage.

Principles (3m W21, S22, S25; 7m S17)

Participation and ownership	Village identifies needs, plans, runs it (CBR committee, gram panchayat)
Inclusion, equal opportunity	Ordinary schools, jobs, festivals, decisions; no discrimination
Empowerment, self-advocacy	Self-help groups, disabled people's organisations
Local resources, appropriate technology	Aids by the village carpenter or blacksmith; household objects for exercise
Multisectoral	Health, education, welfare, livelihood, panchayat together
Integration	Through PHC, anganwadis, schools, not separate systems
Referral	Two-way with PHC, district hospital, District Disability Rehabilitation Centre (DDRC), national institutes
Sustainability	Low cost, local people, community funds; lasts after outside support ends
Gender, children	Equal attention to women and girls; children's evolving abilities

CBR matrix (WHO 2010)

Five components × five elements; a programme picks what it needs. Draw as **five columns**.

- **Health:** promotion, prevention, medical care, rehabilitation, assistive devices.
- **Education:** early childhood, primary, secondary and higher, non-formal, lifelong.
- **Livelihood:** skills, self-employment, wage jobs, financial services, social protection.
- **Social:** personal assistance, relationships, marriage and family, culture and arts, recreation and sport, justice.
- **Empowerment:** advocacy, community mobilisation, political participation, self-help groups, disabled people's organisations.

Team (7m S17) and teamwork benefits (7m S20)

Village: person, family, CBR worker, anganwadi worker, ASHA, teacher, self-help group, gram panchayat, craftsmen. **Block:** PHC and CHC (doctor, ANM), multipurpose rehabilitation worker, block resource centre, RBSK teams. **District:** district hospital, DEIC (0–6), DDRC, PT, OT, prosthetist, speech therapist, psychologist, social worker, NGOs. **State and national:** national institutes, composite regional centres.

Benefits: disability touches health, school, work, social life; few professionals train many local workers (transdisciplinary); shared goals → one plan; early identification, quick referral; follow-up near home; pooled resources → low cost, lasting; attitudes change.

Strategies (7m W19) and approaches (7m S26)

Steps: situation analysis (leaders, resource map, household survey) → plan with community (CBR committee) → train workers and families → implement (home visits, camps, school inclusion, aids, SHGs) → monitor, evaluate, hand over.

- **Urban:** urban health and community centres (clinics, camps); clubs, mahila mandals, social centres (awareness, volunteers, funds); schools (inclusion, screening); **industries** (training, jobs); sports centres; NGOs, CSR funds.
- **Rural:** PHC, rural and district hospitals (screening, treatment, referral); anganwadi workers and ASHAs (identification, home programmes); **local-material aids** by craftsmen (bamboo crutches, wooden walkers, parallel bars, standing frames, CP chairs); panchayat; SHGs; camps.

Approaches (S26). By control: top-down (government-led, set packages; Iran, Viet Nam, DDRC camps): easy to run widely, more money and reach; weak local commitment; **bottom-up** (community, NGO or disabled people's groups; PROJIMO, Mobility India): flexible, owned; fewer resources. Best: government support + local sensitivity (WHO 2010).

By place: home-based (local supervisor teaches family; WHO 1989): reaches homes, cheap; no back-up for aids, referrals fail; **centre-based** (village centre, anganwadi, early intervention centre): back-up, local aids; travel; **outreach** (camps, mobile teams): specialists near home; periodic; **PHC-integrated** (ASHA, anganwadi, RBSK): wide, cheap; stays medical; **community development and rights-based** (matrix, SHGs, disabled people's groups, **twin-track**): tackles poverty and inclusion; complex, little evidence.

Anganwadi worker (3m S17, S22; 7m S26)

ICDS (1975) village worker; centre for ~400–800 people (300–800 tribal, hilly) with one helper; serves children under 6, pregnant and nursing mothers.

- **Identification:** house-to-house survey, growth monitoring, milestone checks.
- **Referral:** PHC, RBSK team, District Early Intervention Centre; certificate and UDID help.
- **Prevention:** immunisation, nutrition, vitamin A, iodised salt, anaemia, antenatal care, birth spacing.
- **Home programme** taught by the physiotherapist; checks aids. **Pre-school inclusion.**
- Counsels mothers, reduces stigma; registers; links to schemes, ASHA, ANM, panchayat.

Physiotherapist in CBR and in a special school (7m W22)

CBR: trainer of workers, teachers, families (assessment, positioning, exercise, transfers, when to refer); assessment at camps, picture home programmes; low-cost local aids; prevention; referral and follow-up; advocacy, supervision.

School for physically handicapped children: screen each child (GMFCS, tone, ROM, posture, gait, hand, self-care), goals in the **individual education plan**; group and individual play-based therapy, home programme; seating (pelvis level, feet supported), standing frame 30–60 min a day; orthoses, walkers, wheelchairs; access (ramps 1:12, rails, toilets); train teachers and parents; adapted PE: WHO 2020, ages 5–17 incl. disability, **60 min a day**, vigorous and strengthening **3 days a week**, as able; prevent contractures, scoliosis, obesity, sores; transition to neighbourhood school, later vocational.

Law and schemes: RPWD Act s16 schools admit without discrimination, access, reasonable accommodation, transport; s17 survey every 5 years, free assistive devices up to 18; s31 **free education 6–18** in neighbourhood or special school. **Samagra Shiksha** (pre-school to XII): assessment camps, aids, escort allowance, girls' stipend, block-level therapy, ramps and toilets, home-based education. UDID, ADIP, RBSK school screening.

Long answer (S22): 6-year-old with CP, village, school

Clues: CP → mobility, positioning, aids; village → PHC, anganwadi; parents want school → **education first**.

Resources (5): family; anganwadi worker, ASHA, ANM; PHC, RBSK, DEIC; village school, block resource centre; **gram panchayat** (ramp, toilet, transport); **carpenter, blacksmith** (CP chair, standing frame, parallel bars, walker); self-help and parents' groups, NGOs; UDID, ADIP, scholarships, concessions.

Plan by matrix (10): assess (CP type, GMFCS, tone, contractures, sitting, walking, hand, communication, vision, hearing; ICF; family goals) → **health:** daily stretches (calf, hamstrings, hip adductors), sitting balance, sit to stand, supported walking, positioning; **standing frame 30–60 min a day** if he cannot stand; AFOs and walker via DEIC or ADIP camp; monthly PT review → **education:** enrol, seat near door and board, adapted chair and table, ramp, accessible toilet, teacher training, peer buddy, extra time, special educator → **social:** play, festivals, stigma counselling, parents' group → **empowerment:** RPWD rights, disabled people's organisation → **livelihood:** family schemes, later vocational → **referral:** orthopaedic review, botulinum toxin or surgery; review every 3 months.

Exam plan

Principles (7): 2004 definition (1) → eight principles with one practical line each (4) → matrix or team by level (2).
Long answer: clues → resources table → plan under the five matrix headings with doses → referral and review.

Draw: CBR matrix as five columns; referral levels village → national institute; bamboo or wooden standing frame and parallel bars. **Underline:** community development, ownership, local resources, appropriate technology, multisectoral, referral, sustainability, twin-track, inclusive education.

Rehabilitation of an amputee in the community

5 papers: below-knee amputee long cases, prosthetic check-outs.

Full notes p. 160

Asked: long cases: 38-year-old teacher found at a village camp 3 weeks after below-knee amputation, functional diagnosis + CBR (S17); 37-year-old truck driver, difficulties + concessions (W18); 32-year-old truck driver, disability evaluation principles + management (W19). **7 marks:** psychosocial issues (W17); prosthetic check-out (S18).

Below-knee prosthesis

Socket: PTB or total surface bearing, soft liner. **Suspension:** supracondylar cuff or wedge, sleeve, pin-lock liner, suction; thigh corset if very short. **Shank:** exoskeletal or endoskeletal (pylon). **Foot:** SACH, single-axis, dynamic; **Jaipur foot** (Sethi, BMVSS, free): cheap, waterproof, allows **squatting, cross-legged sitting, barefoot walking**.

Tolerant: patellar tendon, medial tibial flare, fibular shaft, pretibial muscles, popliteal (lightly). **Relieved:** tibial crest and tubercle, fibular head, distal tibia and fibula, hamstring tendons, patella.

Pre-prosthetic and prosthetic training

Stump: wound, skin check, desensitisation; **figure-of-eight** bandage (tighter distally) or shrinker, day and night → cone. **Prevent knee and hip flexion contracture:** no pillow under the knee, stump board, **prone 20–30 min** a few times a day. Strengthen knee, hip extensors and abductors, trunk, arms; one-leg balance; transfers, crutches. **With prosthesis:** donning, wearing time built up, skin check, weight shift, gait, stairs, slopes, squatting.

Check-out (7m S18)

Before donning	As prescribed; socket smooth; suspension, foot right; stump skin intact
Standing	Equal weight; **pelvis level**; knee slight flexion; foot flat, slight toe-out; **total contact**; patellar bar level; pylon vertical
Sitting	Knee ~**90°** without posterior wall pinching; stays on
Walking	No deviations, no **pistoning**, no pain; equal steps; stairs
After removal	Redness only on tolerant areas, fading in **10–20 min**; no blisters

Deviations: excess knee flexion at heel strike (foot dorsiflexed or back, hard heel); knee stays straight (foot plantarflexed or forward, soft heel); **lateral thrust = foot inset**; medial thrust = outset; early drop-off (short keel); pistoning (loose socket: add socks).

Psychosocial issues (7m W17)

Grief (shock → acceptance); depression, anxiety; body image; **phantom pain**; lost role and income (drivers, farmers); dependence, family and marital strain; **stigma**, beliefs; prosthesis cost. **Help:** counselling, **peer support** from an amputee; early fitting; phantom pain: explanation, desensitisation, mirror therapy, TENS; family; vocational rehab, schemes, sport.

Disability evaluation (W19), difficulties and concessions (W18)

Medical board after maximum improvement; measured function. **Below-knee (BK) up to 8 cm 70%**, lower third **60%**, through-knee 75%, above-knee 80–85%, Syme's 50%; +5% stump unfit, up to +10% complications; two limbs a + b(90 − a) ÷ 90, +10%, max 100%; **40% = benchmark** (disability (p. 22)).

Difficulties: distance, roads, stairs, Indian toilet, energy cost; right leg: driving needs adapted controls; income; repairs. **Concessions:** certificate, UDID; free prosthesis (**ADIP**, BMVSS, DDRC); rail and bus; pension, tax, loans; **4% jobs**, RPWD job protection; accident claims tribunal, employees' compensation, ESI.

ICF and CBR (S17: 38, teacher, BK 3 weeks, remote village)

- **Structure/function:** healing swollen stump, weak deconditioned muscles. **Activity:** walking, Indian toilet, stairs.
- **Participation:** teaching, village life. **Environment:** remote, no prosthetic centre. **Personal:** 38, teacher.

CBR: family and CBR worker taught bandaging, prone lying, exercises; worker weekly, PT monthly; certificate at camp → DDRC, Jaipur foot or ADIP camp; carpenter: parallel bars, stump board, commode stool; panchayat: ramp, rails; early return to teaching (chair, ground floor); peer support, self-help group; follow-up (TUG) (CBR (p. 26)).

Exam plan

Check-out (7): purpose (1) → static: before donning, standing, sitting (3) → dynamic: walking, deviations (2) → after removal (1). **W19:** prosthesis, gait, FCE vs driving, adapted vehicle or other work (disability management (p. 33)). **Draw:** BK prosthesis; figure-of-eight. **Underline:** PTB, tolerant areas, Jaipur foot, knee flexion contracture, foot inset, peer support, ADIP.

Revised ☐☐☐

Disaster management

3 papers: types, principles, the cycle and triage.

Full notes p. 167

Asked: Long answer (S26): disaster management (cycle and triage) and the role of physiotherapists. **7 marks:** types of disaster and principles of disaster management (W19, S24).

Definition and types (7m W19, S24)

Disaster (Disaster Management Act 2005): a catastrophe, mishap, calamity or grave occurrence, natural or man-made, or by accident or negligence, causing substantial loss of life, human suffering, or damage to property or environment, **beyond the coping capacity** of the affected community. **Hazard** = the threat; **vulnerability** = how easily people are harmed (weak buildings, poverty, disability); **risk = hazard × vulnerability ÷ capacity**.

Natural	Geophysical: earthquakes (Latur 1993, Bhuj 2001), tsunami 2004, landslides. Weather: floods (Kerala 2018), cyclones (Odisha 1999), drought, heat waves. Biological: epidemics (COVID-19)
Man-made	Technological: **Bhopal gas leak 1984**, fires, building collapse, nuclear. Transport and crowds: rail, road, stampedes. Conflict: terrorism, war, riots

Also by onset (sudden or slow) and scale (local, state, national).

Principles (7m W19, S24) and India

Comprehensive (all hazards, phases, people); **progressive** (act before); **risk-driven**; **integrated and coordinated** (one plan and command); **collaborative**; **flexible**; **professional**; **community participation and inclusion** (women, elderly, disabled in planning); **build back better** (Sendai).

DM Act 2005: NDMA (chaired by the **Prime Minister**), State and District authorities, NDRF, National Institute of Disaster Management. National plan 2016; **Sendai Framework 2015–2030**.

Disaster management cycle (S26)

Before: mitigation (earthquake-resistant buildings, flood walls, land-use rules, industrial safety, accessible design, risk maps) → **preparedness** (plans, early warning, stockpiles, first aid and triage training, mock drills, lists of elderly and disabled and their evacuation). **After: response** (search and rescue, triage, first aid, evacuation, field hospitals, shelter, food, water, sanitation, epidemic prevention, psychological first aid) → **recovery** (rehabilitation, reconstruction, livelihoods, mental health, long-term disability support) → feeds back into mitigation.

START triage (S26)

Sorting casualties by urgency so most people survive; under a minute per adult; colour tags. Children: JumpSTART.

Walks → **green**. Not breathing after airway opened → **black**; breathes only after opening → red. **RR over 30** → red. **Refill over 2 s** or no radial pulse → red. Cannot obey commands → red; obeys → **yellow**.

30-2-can do

Red (immediate): airway, severe bleeding, shock, tension pneumothorax. **Yellow** (delayed): closed fractures, stable spinal injury, moderate burns. **Green** (minor): sprains, cuts. **Black**: dead or unsurvivable with resources.

Role of the physiotherapist (S26)

Injuries: fractures, **SCI** (many after earthquakes), traumatic amputation, crush, nerve and head injury, burns; later chronic pain, disability, trauma.

- **Preparedness:** in hospital and community plans; first aid, triage, safe lifting, spinal handling; stock crutches, walkers, wheelchairs, splints; evacuation plans for disabled and elderly; teach safe evacuation.
- **Response:** triage, casualty handling; spinal transfer and positioning; fracture splinting; chest PT, breathing exercises; early mobilisation; pressure-sore prevention; amputation care; burns positioning; aids at discharge to free beds; psychological first aid. **WHO: at least 1 rehab professional per 20 beds** in field hospitals.
- **Recovery:** rehab of SCI, amputation, fractures, nerve and brain injury; prostheses, orthoses, wheelchairs; shelter access; CBR (p. 26); vocational rehab; follow-up.
- **Throughout:** records, team coordination, rescuers' health, advocacy for inclusive plans.

Exam plan (S26 long answer)

Definition and types (2) → **cycle (4)** drawn with activities + NDMA, SDMA, DDMA, NDRF → **triage (4)**: START flowchart + four tags with examples → **PT role (5)** by phase + WHO 1 per 20 beds. **7 marks:** definition, types with Indian examples, eight principles, DM Act. **Draw:** cycle; START flowchart. **Underline:** beyond the coping capacity, hazard and vulnerability, mitigation, preparedness, response, recovery, Sendai, START, build back better.

Revised ☐☐☐

Industrial health: models, the team and the worker care spectrum

3 papers: industrial therapy, role of the physiotherapist, pre-placement.

Full notes p. 174

Asked: 3 of 20 papers. **Long answer (S22):** define industrial therapy (2), industrial rehabilitation team (3), job analysis (7) and its purpose (3). **7 marks:** role of the PT in industrial health (W19); pre-placement worker care spectrum (S25).

Definitions and laws (2m of S22)

Occupational health (ILO/WHO 1950, revised 1995): highest degree of physical, mental and social well-being of workers; prevent ill health from working conditions; protect from risks; **adaptation of work to people and of each person to their job**.

Industrial therapy: PT (with other disciplines) applied to workers and workplaces: **prevent work injury**, treat disorders **close to the job**, **return injured workers** to safe, productive work. India: mostly informal workers, little protection.

Laws: Factories Act 1948 (safety, welfare, notifiable diseases); **ESI Act 1948** (care and cash for sickness, maternity, injury, disablement); Employees' Compensation Act 1923; Mines Act 1952; Occupational Safety, Health and Working Conditions Code 2020.

Models of industrial therapy

	Traditional medical	Industrial
Focus	Injury, symptoms	**Function in the job**, prevention
Place	Hospital, away from work	At or near work
Directs	Doctor; patient passive	Team; employer, insurer as **partners**
Return	Off until "fully healed"	**Early**, modified or graded duties
Treatment	Passive, generic exercise	Active, job-specific, ergonomic change
Outcome	Pain, range, strength	Return to work, days lost, cost
Prevention	Little	Central

Industrial rehabilitation team (3m of S22)

Worker and family (active partner, reports early); **employer, supervisor** (modified duties, workplace change); **occupational physician** (diagnosis, fitness certificate, surveillance); **PT** (FCE, work conditioning and hardening, ergonomics); **OT** (hand, ADL, job simulation); **ergonomist, safety officer** (redesign, PPE); **psychologist** (fear of re-injury); **vocational counsellor** (retraining); **case manager, insurer (ESIC), union**.

Job analysis (7 + 3 of S22): description, demand analysis, task analysis (ability management (p. 31)). Purpose: redesign, pre-placement, FCE matching, work hardening, light duty, training, compensation.

Role of the PT in industry (7m W19)

- **Pre-placement** screening; **job analysis and ergonomics** (REBA, RULA, NIOSH; workstation redesign).
- **Prevention, education:** back school, safe lifting, micro-breaks, workstation exercises, warm-up before heavy shifts.
- **Health promotion:** employee fitness, NCD risk, stress. **Early care:** on-site clinic, first aid.
- **Disability management:** FCE, work conditioning, hardening, graded return. **Hazard advice:** vibration, handling, standing, PPE.
- **Records:** injury data, cost–benefit. **Inclusion:** older and disabled workers, **reasonable accommodation** (RPWD Act).

Pre-placement worker care spectrum (7m S25)

Spectrum: pre-placement → ability management → (injury) → disability management → return to work. **Post-offer:** after a job offer, abilities tested against the **essential demands** of the job. **Aims:** safe placement; training, aid or workplace needs; **baseline**; fewer injuries and costs.

Components: job demand analysis; health questionnaire and exam; musculoskeletal screen; **job-matched tests** (lifting, carrying, push-pull, positional tolerance, grip, aerobic). **Result:** fit; fit with adjustments; temporarily unfit; report in **functional terms**. **Rules: job-related, standardised**, same for all; no unfair exclusion (RPWD Act 2016); confidential.

Exam plan

Role (7): definition, industrial model (1) → roles (5) → outcome (1): fewer injuries, days lost. **Draw:** worker care spectrum; two-model table. **Underline:** adaptation of work to people, early return, pre-placement, reasonable accommodation, ESI Act.

Ability management: job analysis, ergonomics and injury prevention

10 papers: job and task analysis, ergonomic evaluation, REBA, workstation exercises.

Full notes p. 179

Asked: 10 of 20 papers, always short answers. **7 marks:** importance of ergonomic evaluation at the workplace (S16), importance of task analysis (S19), REBA (W19), cognitive ergonomics (S21), purpose and principles of ergonomics (S22), define ergonomics and one scale (S23), workstation exercises for a sedentary worker (S25). **3 marks:** workplace health promotion (S18), ergonomic PT evaluation (S19), employee fitness programme (W19), workstation exercises (S20), job demand analysis (S21), job analysis (S26). Also job analysis in the S22 long answer (industrial health (p. 30)).

Worker care spectrum

Pre-placement → **ability management** (before injury: job analysis, ergonomic evaluation, injury prevention, employee fitness) → *injury* → **disability management** (acute care, FCE, work conditioning, work hardening; disability management (p. 33)) → **return to work** → back to ergonomics and prevention.

Pre-placement: after a job offer, abilities matched with job demands; baseline; no unfair exclusion of PWD (RPWD Act 2016). **Ability management:** keeping healthy workers healthy and productive by **fitting the job to the worker and the worker to the job**.

Job analysis (3m S26; S21)

Job description	Title, purpose, duties, tools, materials, hours and shifts, breaks, workplace, supervision, pay (piece rate or salary)
Job demand analysis	Physical, mental and environmental demands and **how often** each occurs
Task analysis	Each task in steps: **posture, force, repetition, duration, tools, environment**; risk factors found

Job demand analysis (S21): physical (lift, carry, push, pull with weights and distances; sit, stand, walk time; bend, kneel, squat, climb, reach above shoulder; grip, fine work, repetition; vibration); **mental and sensory** (attention, memory, decisions, vision, hearing, deadlines); **environment** (heat, cold, noise, light, dust, fumes, heights, wet floors, shifts).

Frequency (Dictionary of Occupational Titles): occasionally up to a third of the day; **frequently** a third to two-thirds; **constantly** more than two-thirds.

Demand level	Lifting
Sedentary	Up to 10 lb (about 4.5 kg) occasionally; mostly sitting
Light	Up to 20 lb (9 kg) occasionally, 10 lb frequently
Medium	Up to **50 lb (23 kg)** occasionally, 25 lb (11 kg) frequently
Heavy	Up to 100 lb (45 kg) occasionally, 50 lb frequently
Very heavy	Over 100 lb occasionally, over 50 lb frequently

Importance of task analysis (7m S19)

- Finds the **exact steps and postures** causing pain, so the fix is specific; basis for **ergonomic redesign**.
- **Matches job to worker** in pre-placement and return to work (compare with FCE).
- Designs **work hardening** (job simulation); shows **light duties**; guides rotation.
- Basis for safe-method training, fair production rates, compensation decisions.
- Example: lifting bags into a truck: bending, twisting, 50 kg, 3 ft height.

Ergonomics: definition, purpose, principles (7m S22, S23)

IEA (2000): "the scientific discipline concerned with the understanding of interactions among humans and other elements of a system"; the profession applies design to **optimise human well-being and overall system performance**. Simply: **fitting the job to the person**.

Physical	Anatomy, anthropometry, biomechanics: postures, handling, repetition, MSDs, layout
Cognitive	Perception, memory, reasoning: workload, decisions, human–computer interaction, error, stress
Organisational	Shift design, teamwork, communication, pace, participation

Purpose: prevent MSDs and accidents; less fatigue and error; comfort, productivity, quality; include older and disabled workers.

Principles: neutral postures (spine in its curves, elbows by side, wrists straight, neck upright); less force (lighter loads, aids, good grips); easy reach; **work at elbow height** (lower for heavy, higher for fine); less repetition and static load; breaks and rotation; minimise contact stress and vibration; space to move, sit–stand; comfortable light, temperature, noise; clear displays and controls; **participatory ergonomics**.

Cognitive ergonomics (7m S21)

- **Mental workload:** overload (errors, stress) or underload (boredom, lapses); balance tasks and breaks.
- **Vigilance:** monotonous monitoring, night shifts, fatigue reduce it.
- **Displays and controls:** clear labels, logical layout, warnings that stand out, controls that move as expected.
- **Human error:** hard to make, easy to correct (checklists, forcing functions). Computer work: simple screens, fewer interruptions; training, decision support, stress management.
- Examples: control rooms, drivers, pilots, surgeons, call centres.

Ergonomic PT evaluation (3m S19), importance (7m S16)

1. Meet management and workers; injury and absence records; **Nordic Musculoskeletal Questionnaire**.
2. Walk-through and video; job and task analysis.
3. Measure workstation (heights, reaches, seat, screen), loads, forces, frequency, environment.
4. Score risk: **RULA, REBA, NIOSH lifting equation, Strain Index**.
5. Recommend from the **hierarchy of controls**: redesign, aids, layout, breaks, rotation, training, exercise → **re-evaluate**.

Importance: finds risk **before injury**; prevents MSDs, accidents, absence; comfort, productivity; lower compensation; **objective scores** to prioritise and show change; return and inclusion of workers; legal duty.

REBA (7m W19; scale S23)

Hignett and McAtamney (2000): whole-body postures in **unpredictable or varied tasks** (health care, service, industry). **Group A:** trunk, neck, legs + load → score A. **Group B:** upper arm, lower arm, wrist + **coupling** (grip) → score B. A + B → score C + **activity score** (static over a minute, repeated small actions, rapid large changes) → **REBA 1–15**.

1	Negligible: none needed
2–3	Low: may be needed
4–7	Medium: needed
8–10	High: needed soon
11–15	**Very high: needed now**

RULA: upper-limb version for seated and standing arm work (computer, assembly), **1–7**.

Prevention, employee fitness (3m W19), health promotion (3m S18)

Injury prevention: redesign first, then safe methods, training, PPE; early reporting and PT (stage 1 CTD reversible); warm-up before heavy shifts, micro-breaks, rotation.

Employee fitness programme: needs assessment, management support, worker committee → health risk appraisal, fitness tests (BP, BMI, waist, aerobic, strength, flexibility) → classes or gym, walking groups, stretch breaks, sports; education (back care, nutrition, smoking, stress, sleep) → incentives, convenient timing, family → **evaluate** participation, injuries, absence. **Benefits:** fewer injuries and sick days, morale, productivity, lower costs.

Workplace health promotion (Luxembourg Declaration 1997): "the combined efforts of employers, employees and society to improve the health and well-being of people at work"; via work organisation, participation, personal development. Examples: ergonomic workplaces, health checks, healthy canteens, **tobacco-free workplace**, stress management, flexible hours.

Workstation exercises (3m S20; 7m S25)

Sedentary worker: **5–10 repetitions, every 30–60 minutes**; pair with the desk set-up (mechanical stress (p. 37)).

Neck	Chin tucks; side bending, turning, **hold 10 s**
Shoulders	Rolls, shrugs; squeeze blades back; reach overhead
Mid back	Extension over chair back; seated twists; side bends
Wrists, hands	Flexor and extensor stretches; fist and spread; tendon gliding
Low back	Standing back bends; pelvic tilts in sitting; stand and walk
Legs	Ankle pumps, circles; knee straightening; heel raises; marching
Eyes	Look into the distance; blink; palming

Exam plan

REBA or any scale (7): IEA definition, domains (1) → tool: authors, groups A and B, load, coupling, activity, action table (4) → use: when, example job, scores guide change (2). **Task analysis (7):** three parts of job analysis → uses → example.

Draw: worker care spectrum; work heights (precision above elbow, light at elbow, heavy below). **Underline:** job demand analysis, task analysis, occasionally / frequently / constantly, physical, cognitive, organisational, neutral posture, REBA 1–15, workplace health promotion.

Revised ☐☐☐

Disability management: functional capacity evaluation, work conditioning and work hardening

10 papers: FCE, work conditioning, work hardening.

Full notes p. 186

Asked: 10 of 20 papers. **Long cases:** 32-year-old sugarcane worker with neck and arm pain: job task analysis, work hardening and conditioning (W18); 54-year-old supervisor after fracture fixation: functional capacity assessment, work conditioning and hardening (W20). **7 marks:** FCE (W16, S17, S19); work hardening with two examples (W18). **3 marks:** job simulation (S16); work conditioning (S17, S22); define work hardening and conditioning (W21, W22); define functional capacity assessment (W22, W25).

Disability management and acute care

Disability management: the "after injury" half of the worker care spectrum (p. 31): a planned process by employer, worker and health team to **limit the effect of an injury on work** and bring the worker back **safely and early**.

Steps: early reporting → acute care → case management, contact with employer → **FCE** → **work conditioning** → **work hardening** → modified or transitional duties → full return → prevent recurrence.

Acute care: diagnosis and treatment; pain and swelling control (rest from aggravating tasks, ice, elevation, medication, electrotherapy); **early graded movement**, no long bed rest; education, reassurance, **expected return date set early**; employer contact on light duties, keep the worker connected; screen **yellow flags** (fear of movement, low mood, job dissatisfaction).

Functional capacity evaluation (7m W16, S17, S19; 3m W22, W25)

Definition: a **standardised set of tests** measuring a person's **safe maximum ability** to perform work-related physical tasks, compared with the demands of a job or a **physical demand level**. **Uses:** return-to-work decisions, job matching; planning conditioning and hardening; modified duties; disability rating, compensation; pre-placement; progress.

Intake	History, job description, symptoms, self-rated abilities, medical clearance, consent
Musculoskeletal screen	Posture, ROM, strength, sensation, HR and BP
Material handling	Lift **floor to waist, waist to shoulder, overhead**; carry; push, pull; weights raised to a safe maximum
Positional tolerance	Sit, stand, walk, stoop, kneel, crouch, reach, stairs and ladders, balance
Hand function	Grip (dynamometer), pinch, dexterity
Aerobic	Step or treadmill test
Effort, consistency	Repeated trials, HR response, observed body mechanics
Report	Safe abilities, demand level, comparison with job, **projection to an 8-hour day**, recommendations

Approaches: kinesiophysical (Isernhagen Work System, 28 tests; body mechanics decide the safe maximum); **psychophysical** (worker judges an acceptable load). Several hours, often over **1–2 days**.

Work conditioning vs work hardening (3m S17, S22, W21, W22)

	Work conditioning	Work hardening
Definition	Intensive, work-related, goal-oriented conditioning to restore systemic, neuromusculoskeletal and cardiopulmonary function (strength, endurance, mobility, motor control, aerobic capacity) for return to work	Highly structured, goal-oriented, individualised, **interdisciplinary** programme using **real or simulated work tasks** to restore physical, behavioural and vocational functions for a **specific job**
Team	Usually **one discipline** (PT)	PT, OT, psychologist, vocational counsellor, case manager
Focus	Physical capacity	Job tasks, productivity, safety, tolerance, work behaviours, pain coping
Hours	Part of a day	Builds to a **full working day**
Entry	After acute care, physical deficit	Capacity nearly enough; tolerance, confidence or behaviour limit return

Content, examples (7m W18), job simulation (3m S16)

Work conditioning: aerobic (treadmill, cycle, walking) **20–40 min**; progressive strengthening of the injured area and whole body (trunk, legs, grip); flexibility, motor control, core stability; basic body mechanics and lifting; education.

Work hardening: job simulation from the job analysis, graded in weight, height, distance, repetitions, time; conditioning continued; body mechanics and pacing; ergonomic changes for the real job; pain, stress and fear-of-movement counselling; work habits (timekeeping, instructions); re-measure against job demands; plan graded return.

- **Warehouse loader, low back strain:** lift crates floor → waist → shoulder, **10 kg to 25 kg**; carry 20 m; push a loaded trolley; load a mock truck bed; build to the full shift's lifts.
- **Data-entry clerk** after carpal tunnel release or wrist fracture: typing and mouse **15 to 60 min**; filing, lifting files; dexterity; workstation set-up, micro-breaks; build to a full day.

Job simulation: reproducing the real tasks (same tools, weights, heights, postures, pace) in the clinic or a mock workstation, to **train** (work hardening) and **test** readiness (FCE). Examples: work simulators with interchangeable handles, Valpar work samples, mock construction, assembly or office stations.

Return to work: same job full duty → same job with changes → modified or light (transitional) duty → different job, same employer → retraining. **Graded:** part days, lighter tasks, increase weekly. Prevent recurrence: ergonomics, home exercise, fitness, early reporting.

Long case W18: job task analysis (sugarcane worker, 32, 65 kg)

Cutting cane	Stooping, squatting for hours, forceful sickle strokes, heat → sustained neck flexion, repetitive shoulder and elbow load
Bundling, carrying	Heavy bundles on head or shoulder, uneven fields → **axial cervical compression**, one-sided load
Loading	Lifting above shoulder, twisting, climbing → overhead lifting with neck extension
Mill work	Feeding crushers, sugar bags, vibration, noise, heat, shifts
Organisation	**Piece rate**, long harvest days, few breaks → fatigue

Assess: neck range, neuro screen (myotomes, reflexes, sensation, Spurling's, upper limb tension tests), deep neck flexor, scapular and grip strength, pain scale, Neck Disability Index; FCE: floor-to-waist and waist-to-shoulder lifts, carrying, overhead and repetitive arm work, stooping; compare with job (**very heavy work**).

ICF and plan (W18)

- **Structure/function:** 6 months mechanical neck pain with arm referral (rule out radiculopathy); reduced neck mobility; weak deep neck and scapular muscles; low work tolerance.
- **Activity:** head and shoulder loading, stooped cutting, overhead loading. **Participation:** cannot manage full harvesting or loading work.
- **Environment:** head-loading, piece rate, heat, few breaks. **Personal:** young, light build carrying heavy loads; deconditioning and fear likely.

Work conditioning (weeks 1–3): neck and thoracic mobility, nerve gliding, **10 reps twice a day** (heat or TENS first if needed); deep neck flexors, isometrics, scapular retraction and lower trapezius with bands **3 × 10** → resisted shoulder work; aerobic **20–30 min**, trunk, hip, leg, grip strength; hip-hinge and squat lifting with light loads.

Work hardening (weeks 3–8): stooped cutting with a weighted handle; carry bundles **10 kg rising to job load**; load onto a raised platform; longer towards a full shift. **Avoid head-loading**: padded yoke on both shoulders, cart or wheelbarrow, share loads, keep close. Micro-breaks **every 30–60 min**, water in heat; pain education, graded exposure, full-day attendance. **Return:** cutting without carrying first, then full work; talk to the contractor on load size and carts.

Practice case W20 (supervisor, 54, after fracture fixation)

FCE: intake, medical clearance, **fracture healing limits**, ROM, strength, walking and stairs, standing and sitting tolerance, lifting and carrying within limits, hand function, aerobic test, report. **Work conditioning:** strength, mobility, gait, balance, aerobic. **Work hardening:** simulated floor rounds, stairs with files, desk work, building to full days; modified duties first.

Limits and cautions

FCE is affected by **pain, fear and motivation**, predicts return only modestly, needs a trained evaluator and standard protocol. Avoid long bed rest. Yellow flags delay recovery. Test within fracture healing limits; safe stopping criteria in lift tests.

Exam plan

FCE (7): definition and uses (2) → components (4) → approaches and limits (1). **Long case:** job task analysis table → functional diagnosis → FCE vs job demands → work conditioning (doses) → work hardening (job simulation) → graded return, prevention.

Draw: worker care spectrum with the after-injury steps; WC vs WH table. **Underline:** early return to work, FCE, material handling, positional tolerance, physical demand level, work conditioning: physical, one discipline; work hardening: job-specific, interdisciplinary; job simulation, graded return.

Revised ☐☐☐

Environmental stress at work: physical, chemical and mechanical hazards

12 papers: vibration, pneumoconiosis, cumulative trauma disorders.

Full notes p. 193

Asked: 12 of 20 papers. **Long cases:** automobile worker with vibration and numb hands (S16), sand-industry worker (S16), clerk with carpal tunnel syndrome (W17), coal miner (S21). **7 marks:** workplace hazards and risk assessment tools (S18), physical agents (W20), factors in industrial injury (S22), CTD with one example (W22, S24). **3 marks:** causative factors for CTD (W16), byssinosis (S18), asbestosis (W18), three hazards of chemical agents (S19), sequelae of pneumoconiosis (W20), types of hazards (W25).

Types of occupational hazards (3m W25)

Physical (heat, cold, light, noise, vibration, radiation, air pressure); **chemical** (dusts, fumes, solvents, metals, pesticides); **biological** (TB, hepatitis B, anthrax, leptospirosis); **mechanical and ergonomic** (machines, falls, lifting, posture: mechanical stress (p. 37)); **psychosocial** (job stress, shifts, monotony, harassment).

Physical agents (7m W20)

Agent	Effects → prevention
Heat	Cramps, syncope, exhaustion, **heat stroke** → shielding, ventilation, acclimatisation, water and salt, cooler hours
Cold	Chilblains, frostbite, trench foot, hypothermia → insulated clothing, warm breaks
Light	Eye strain, accidents, **miner's nystagmus** → even lighting, no glare, task lights
Noise	Hearing loss **first at 4 kHz**, tinnitus, stress → enclosures, **8 h at 90 dBA** (Indian factory rules), plugs, muffs, hearing tests
Vibration	Hand-arm: white finger, numbness, weak grip; whole-body: low back pain → EU limits: hand-arm action **2.5**, limit **5 m/s²**; whole-body 0.5, 1.15 m/s²
UV	Welder's flash, sunburn, cataract, skin cancer → shields, goggles, screens, clothing
Ionising	Burns, sickness, cataract, cancer, infertility → **time, distance, shielding**, dosimeter; **20 mSv a year** over 5 years

Chemical agents (3m S19) and pneumoconioses (S18, W18)

Three routes: local action on skin, eyes (dermatitis, burns, ulcers, cancer); **inhalation** (pneumoconiosis, asthma, CO, lead, solvents); **ingestion** from hands, food, smoking at work (lead, mercury). **Pneumoconiosis:** lung fibrosis from inhaled mineral or organic dust.

Silicosis	Silica: quarrying, mining, **sand industry**, glass, pottery, slate-pencil, agate; upper-lobe nodular fibrosis, progressive massive fibrosis (PMF); **silico-tuberculosis**; no cure
Coal workers'	Underground mining; nodules → PMF; chronic bronchitis; Caplan's (with rheumatoid arthritis)
Asbestosis (W18)	Mining, cement sheets, insulation, brake linings, ship-breaking; **lower-lobe** fibrosis, pleural plaques, clubbing; lung cancer (more with smoking), **mesothelioma** decades later
Byssinosis (S18)	Cotton, flax, hemp: carding and blowing rooms; **"Monday chest tightness"** on the first day back, later chronic obstruction

Schilling grades: ½ occasional tightness on the first working day; **1** every first day; **2** first and other days; **3** grade 2 with permanent incapacity.

Sequelae of pneumoconiosis (3m W20)

Progressive fibrosis, restrictive disease, PMF; **tuberculosis** (silica); chronic bronchitis, obstruction; **pulmonary hypertension, cor pulmonale**, respiratory failure; lung cancer, mesothelioma, pneumothorax; disability, loss of work and income, early death.

PT and prevention for dust disease

PT: breathing control, pursed-lip breathing; ACBT if sputum; thoracic mobility; graded exercise (pulmonary rehabilitation); energy conservation; oxygen as prescribed; quit smoking; vaccines.

Prevention: substitute (asbestos bans); **wet methods**, enclosure, exhaust ventilation; fitted respirators (N95 or better); pre-placement and periodic **chest X-ray (ILO) and spirometry**; TB screening in silica workers; remove from exposure; notifiable (Factories Act); compensation.

CTD (3m W16; 7m W22, S24) and carpal tunnel syndrome (CTS)

CTD: injury of muscles, tendons, nerves, joints from **repeated small strains without enough recovery**. **Work factors:** repetition, force, awkward or static posture, vibration, contact pressure, cold, no breaks, piece rate. **Personal:** age, female, pregnancy, menopause, diabetes, hypothyroidism, rheumatoid arthritis, obesity, previous injury, low fitness, smoking, stress. **Examples:** CTS, de Quervain's, trigger finger, lateral epicondylitis, rotator cuff, tension neck, hand-arm vibration.

Stage 1	Ache during work, settles overnight; work normal; **reversible**
Stage 2	Pain early in shift, lasts overnight; sleep disturbed; capacity reduced
Stage 3	Pain at rest and with light tasks; cannot do light duties; months to years

CTS: median nerve under the flexor retinaculum; tingling thumb, index, middle, half ring, **worse at night**, shaking helps; later **thenar wasting** (abductor pollicis brevis). **Tests: Phalen's** (wrist flexion 60 s), **Tinel's**, carpal compression; nerve conduction. **Treat:** night splint in neutral; nerve and tendon gliding; ergonomics, breaks; injection; surgery if wasting, constant numbness or failed care.

Hand-arm vibration syndrome (HAVS)

Grinders, drills, riveters. **Vascular:** cold-triggered **white finger**, then blue, red. **Sensorineural:** tingling, numbness, poor dexterity, often CTS. **Musculoskeletal:** weak grip, pain. **Stockholm scale:** vascular 0–4 (4 = skin changes at tips); sensorineural 0SN–3SN (3SN = reduced discrimination or dexterity).

Gear (3m of S16): low-vibration tools, anti-vibration handles; **tool balancers**, suspension; maintenance; **warm, dry gloves** (anti-vibration gloves cut little); vibration meters, time limits; warm workplace; no smoking.

Industrial injury (7m S22) and risk assessment (7m S18)
Factors: worker (fatigue, no training, haste, alcohol, no protection); **machine** (unguarded, faulty); **environment** (light, noise, heat, slippery floors, height); **organisation** (long hours, night shifts, pressure, no rules or PPE); **ergonomic** (heavy lifting, posture, repetition).

Risk assessment: identify hazards → who is harmed → evaluate (**likelihood × severity**) → record and act → review. **Tools: RULA** 1–7 (7 change immediately); **REBA** 1–15 (11+ very high); **NIOSH** lifting equation (**23 kg** ideal limit, lifting index); Strain Index, OCRA; OWAS, QEC; Nordic Questionnaire.

Hierarchy of controls: elimination → substitution → engineering (guards, wet methods, hoists) → administrative (rotation, breaks, training) → **PPE last**.

ICF (S16 case: 45, automobile worker, vibration, no protection)

- **Structure/function:** HAVS ± bilateral CTS; tingling, numbness, pain, poor two-point discrimination, weak grip and pinch, possible blanching; Phalen's, Tinel's positive.
- **Activity:** holding tools, fine tasks, carrying. **Participation:** full factory work; job and income at risk.
- **Environment:** vibrating tools, no PPE, cold workshop (barriers); safety officer, ESI (facilitators). **Personal:** earning member, possibly smoker.

Short-term goals (4–6 weeks): less pain and tingling, no blanching at work; better grip, pinch, dexterity; exposure **below the action value**; knows warm-hands routine and warning signs.

- **Remove exposure** (main treatment): reduce tool work, rotate; refer for nerve conduction, vascular tests.
- **Nerve gliding** (median, ulnar) **5–10 reps, 3 times a day**; night splint in neutral.
- **Pain:** warm (not hot) soaks, TENS. **Grip:** putty, gripper, opposition, pegboard, coins **10–15 min twice a day**.
- **Blanching:** keep warm, gloves, no smoking, warm-up exercises. **Numb skin:** protect, check daily; sensory re-education.

Practice cases (S16 sand, S21 coal, W17 clerk)
Sand worker: silicosis, rule out TB; low exercise tolerance; dust PT and prevention. **Coal miner:** task analysis: stooping, kneeling, crawling; drilling (dust, vibration); shovelling; heat, poor light, noise, shifts; roof falls, gas → coal workers' pneumoconiosis with chronic bronchitis → wet drilling, sprays, ventilation, respirators, periodic X-ray and spirometry, rotation, quit smoking, Mines Act. **Clerk with thenar wasting:** refer for nerve conduction, surgical opinion; breaks, nerve gliding; keyboard at elbow height, neutral wrists, mouse close; **cock-up splint, wrist neutral**, night and flares.

Exam plan
CTD (7): definition, factors (2) → stages, examples (1) → CTS in detail (3) → prevention (1). **Any hazard:** effect → PT → prevention from the top of the hierarchy.

Draw: hierarchy of controls; carpal tunnel cross-section (median nerve, tendons). **Underline:** 4 kHz, white finger, silico-tuberculosis, Monday chest tightness, mesothelioma, repetition, force, posture, Phalen's, Tinel's, PPE last.

Mechanical and psychological stress at work

17 papers, with 16 long cases: computer users, drivers, standing workers, labourers.

Full notes p. 203

Asked: 17 of 20 papers. **16 long cases:** desk workers (W16, S21, W21, W25, S26), bus drivers (S17, W17), watchman (S23), traffic police (S24), labourer (W22), squatting worker (S19), teacher (S19, S24), goldsmith (S20, W24), tailor (S25). **7 marks:** housemaid (S16), psychological hazards and job stress (W17, S18, W18), physiotherapists (S19), PT in stress management (S20, W21, W24), prolonged standing (S21), low back pain in industry (W22, W25), surgeons (S26). **3 marks:** air hostess (W18), shift work (W21), truck drivers (S24).

How work causes pain

Cumulative microtrauma, not one injury (CTD (p. 35)).
Physical: awkward or static posture held long, repetition, force, vibration, contact pressure, cold, too little rest.
Personal: low fitness, obesity, previous injury, smoking, age.
Psychosocial: high demand, little control, deadlines, monotony, poor support: more muscle tension, pain persists.

PT method: job and task analysis → risk factors → treat impairments → **change the workstation or way of working** → self-management, fitness.

Desk and computer workers (long cases)

Problems: neck and shoulder pain with **forward head, rounded shoulders**; thoracic stiffness; low back pain from slumping; wrist symptoms (carpal tunnel); eye strain; weight gain.

- **Screen:** in front, **arm's length (50–100 cm)**, top line **at or slightly below eye level**; laptop on a stand, separate keyboard and mouse.
- **Chair:** feet flat, thighs level; **lumbar support**. **Keyboard, mouse:** close, **elbow height**, wrists straight.
- **Headset** for calls; **movement break every 30–60 min**.
- **PT:** deep neck flexor, scapular retractor, lower trapezius strengthening + general fitness (beats ergonomics alone: Chen 2018); thoracic mobility, pectoral stretch; workstation exercises (p. 31); heat, TENS **only as a bridge to exercise**; weight control.

Drivers (S17, W17 LAQ; truck drivers 3m S24)

Prolonged fixed sitting	Back, neck pain, poor leg circulation → **break every 2 hours**: walk, stretch
Whole-body vibration	Low back pain, disc degeneration → suspended, maintained seat; limit hours (EU action **0.5**, limit **1.15 m/s²** over 8 h)
Poor seat, reach	Twisting, neck strain → knees slightly bent at pedals; backrest slightly reclined, lumbar support; headrest; mirrors set
Loading, climbing	Back, knee strain, falls → handholds, steps, do not jump down, lift with help
Long hours, nights	Fatigue, obesity, accidents → planned sleep, healthy food, exercise

Psychological (W17): stress, irritability, anxiety, poor sleep, loneliness → planned rest, relaxation, fitness. **Cervical spondylosis (S17):** neck isometrics, deep neck flexors, thoracic mobility; headrest, lumbar roll; stretch breaks.

Prolonged standing (7m S21; watchman S23, traffic police S24)

Effects (Waters and Dick 2015): low back pain, leg pain, swelling, fatigue, **varicose veins**, heel pain. **Why:** static load, little calf pumping, hard floors, poor shoes.

Prevention: change position and walk (most effective); sit–stand or leaning stool; **anti-fatigue mats**; cushioned shoes, insoles; footrest to alternate feet; compression stockings if swelling; **calf raises, ankle pumps** in the shift.

Surgeons (7m S26): Evaluate: Nordic questionnaire → video → measure set-up → **RULA, REBA** → change → re-evaluate. **Open:** hands near **elbow height, elbows 90°**; light loupes angled down. **Laparoscopic:** table **0.7–0.8 × elbow height**; monitor **straight ahead**, top at eye level; handles for a straight wrist. **Microsurgery:** sit, forearms supported, eyepieces for a neutral neck. **Micro-breaks 1.5–2 min every 20–40 min**; mat; ratchet instruments.

Manual handling, low back pain in industry (7m W22, W25)

Factors: heavy, frequent, awkward lifting (bend, twist), far from body or from the floor; whole-body vibration; long sitting or standing; slips; low fitness, smoking, obesity; dissatisfaction, stress.

Revised NIOSH equation: 23 kg limit in ideal conditions, reduced for distance, height, travel, twist, frequency, grip; 50 kg bags far exceed it, so **the task must change**.
Prevention: redesign first (lighter bags, trolleys, hoists, conveyors, waist-height loads, team lifts) → training, fitness.
Current: lifting-technique training alone did not prevent back pain (Verbeek 2011).

W22 labourer (50 kg bags, bidis): education, graded activity, core, hips, hamstrings, quit smoking; **25 kg bags or two-person lifts**, trolley, waist-height loading. **Squatting (S19):** low stools, kneeling pads, rotate; quads, hips.

Other occupations: housemaid (7m S16), physiotherapists (7m S19), air hostess (3m W18)

Goldsmith (S20, W24): neck flexion over fine work, pinch, poor light → raise work, magnifier, light, micro-breaks, nerve gliding. **Tailor (S25):** bent sitting, pedal → back support, machine at elbow height, walk breaks. **Teacher (S19, S24):** arms raised at board → write at chest to eye height, projector, alternate sit and stand.

Housemaid: pain knees 38.2%, **low back 27.5%** (Bengaluru); low back risk: **over 10 years** in job, distress. Short jhadu → **long-handled broom**; hand mopping → long mop, knee pad; squat washing → **low stool (paat)**, raised platform; water → **half-filled buckets**, two small loads close, turn with feet. PT: stay active; bracing, bridging, sit-to-stand (8–12 reps, 2–3 days a week); stretches; weight control. **Current (NICE):** no belts, TENS, ultrasound, IFT.

Physiotherapists: lifetime **55–91%**, low back commonest; top cause **transfers**; thumbs with manual therapy. Patient limit about **16 kg (35 lb)** → **hoists, slide sheets**; adjustable plinth **5–10 cm below elbow** light work, **20–40 cm** heavy; straight wrists; breaks, rotation, fitness; body mechanics alone does not prevent injury.

Air hostess: back injuries mostly from **turbulence, overhead bins, trolleys** → no heavy overhead lifts, bags under seat; push trolleys with both hands, close; sit belted in turbulence; shoes, calf raises, sleep plan.

Job stress (7m W17, S18, W18) and shift work (3m W21)

Causes: demand–control (high demand, little control); **effort–reward imbalance**; monotony, dissatisfaction; deadline anxiety; long hours, shifts; poor relationships, harassment, insecurity. **Effects:** muscle tension, pain, headache, fatigue, poor sleep, high BP, anxiety, depression, errors, absence. **Prevent:** fair workload, control, variety, breaks, support, clear roles.

Shift work: disturbed body clock: poor sleep, fatigue, accidents; digestive, weight, metabolic and heart risk; night shifts **"probably carcinogenic" (IARC 2019)**; family strain. Advice: **forward-rotating shifts**, dark quiet sleep room, regular meals, exercise, short runs of nights.

PT in stress management (7m S20, W21, W24)

Relaxation: Jacobson progressive (tense then release each group); **Laura Mitchell** physiological (move into the opposite of the stress posture); diaphragmatic breathing. **Exercise:** aerobic, workplace stretching, yoga. Ergonomic and task changes; posture, sleep, pacing. **Refer** anxiety or depression; evaluate (pain, absence, stress questionnaires).

ICF (S21 case: 36, computer operator, mother of a 1-year-old)

- **Structure/function:** neck and upper trapezius pain, less range, forward head, weak deep neck flexors and scapular muscles, tight pectorals, poor sleep.
- **Activity:** long typing, looking down, lifting and feeding the child. **Participation:** full working day, childcare, leisure.
- **Environment:** unadjusted workstation, long hours, no breaks (barriers); family, employer willing to adjust (facilitators). **Personal:** working mother, little time: short home exercises.

Short-term (2–4 weeks): pain **6 → 3/10** at work; corrected sitting; workstation adjusted, breaks every 30–60 min; safe lifting and feeding. **Long-term (2–3 months):** pain-free working day; posture held; self-checks set-up; exercise **3 times a week**; no recurrence.

Plan: desk-worker PT and workstation (above); child close, supported feeding; relaxation.

Cautions

Neck pain: rule out radiculopathy. **Tailor (S25) who cannot walk over 15 min:** examine for spinal stenosis or vascular claudication (lumbar extension tests, foot pulses, ankle–brachial index, neurological exam).

Exam plan (15-mark occupational case)

1. **Task analysis or diagnosis (5):** posture, load, hours, tools, environment; ICF; one line.
2. **Goals (5):** short and long term, measurable, linked to work and home.
3. **Management and prevention (5):** pain relief, exercise with doses, workstation, breaks, health promotion, return to work.

Draw: workstation (screen 50–100 cm, top at eye level, elbows 90°, lumbar support); driver's seat; lifting with load close; surgeon at elbow-height table. **Underline:** micro-breaks, whole-body vibration, NIOSH 23 kg, hoists and slide sheets, redesign before training, demand–control.

The project synopsis

25 practical marks: the synopsis you present and defend.

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Asked: not set in theory papers. The synopsis carries **25 marks** in the university practical (with long case 50, journal 5) and **30** in the preliminary; expect a presentation and viva on question, design, sample size, outcomes, statistics and ethics.

What it is and what examiners want

Synopsis: a short, complete plan of a study (what you will ask, why, exactly how), carried out in internship; any area of physiotherapy. **Examiners look for:** clear answerable question, fitting design, defined variables, feasible sample and setting, valid outcomes, sensible analysis, ethics, and that you can explain every choice.

Choosing the question

FINER: Feasible (participants, internship time, skills, money); **I**nteresting; **N**ovel (new population, Indian or rural setting, outcome); **E**thical (consent, approvable); **R**elevant (practice, policy). **PICO:** Population, Intervention, Comparison, Outcome: desk workers with neck pain; deep neck flexor training vs stretching; Neck Disability Index at 4 weeks.

Good topics: prevalence surveys (back pain in bus conductors, postnatal incontinence, falls risk in a seniors' club); correlation (BMI and knee pain); small comparisons of two techniques. **Avoid:** rare conditions, long follow-up, costly equipment, vague titles.

Sections of a synopsis

Title (design, intervention, population, outcome) → **introduction** (problem with numbers, impact, gap) → **need for study** → **review of literature** (**10–20** recent studies by theme, ending with the gap) → **question and hypotheses** (null: no difference; alternative) → **aim and objectives** (one aim; measurable objectives) → **methodology** → **ethics** (ethics committee, written consent in local language, confidentiality, right to withdraw; **CTRI** (Clinical Trials Registry–India) for a trial) → **timeline** (Gantt) and budget → **references** (one style, usually Vancouver) → **annexures** (consent, information sheet, proforma, questionnaires, protocol).

Methodology

Designs: cross-sectional (prevalence: neck pain in IT workers); correlational; case–control (past exposures); cohort (follow forward); quasi-experimental pre–post; **RCT** (random allocation, strongest for cause and effect).

- **Population:** where and who; inclusion and exclusion criteria, measurable, with reasons.
- **Sampling:** probability (simple random, systematic, stratified, cluster) or non-probability (convenience, purposive, snowball); allocation by sealed envelopes or computer list.
- **Sample size:** from a previous study (prevalence or mean difference and SD), **5% significance, 80% power**, + **10–20%** drop-outs; state the source.
- **Variables:** independent (intervention, exposure), dependent (outcomes); control confounders.
- **Outcomes** valid and reliable: VAS, NPRS, goniometer, Neck Disability Index, Oswestry, Berg, TUG, 6MWT, ICIQ, PERFECT.
- **Procedure** repeatable: dose (frequency, intensity, time, type, duration), when and who measures (blinding).

Statistics

Question	Normal	Non-normal
Before vs after, one group	Paired t	Wilcoxon signed-rank
Two independent groups	Unpaired t	Mann–Whitney U
Three or more groups	ANOVA	Kruskal–Wallis
Two categorical	Chi-square	Fisher's exact
Relationship	Pearson	Spearman

Mean and SD (or median and range), percentages; **$p < 0.05$**; name the software.

Presentation and viva

8–10 min: title, guide → problem with one number → gap (3–4 studies) → aim, objectives, hypotheses → design, setting, sample size, criteria → outcomes and study flowchart → analysis, ethics, timeline → expected use.

Asked: why this topic, what is new; why this design, limits; sample size; dependent vs independent; validity and reliability of outcomes; bias (randomisation, blinding); null hypothesis and test; consent, harm, drop-outs; feasibility in internship.

Tips: know every reference; admit limits; carry consent form, proforma, questionnaire; simple slides; write it yourself (copying fails). **Draw:** study flowchart (eligible → excluded → randomised → two groups → outcomes → analysis); Gantt chart. **Underline:** FINER, PICO, null hypothesis, power 80%, valid and reliable, informed consent, CTRI.

Answer writing for the Community paper

The paper, long cases and answer skeletons.

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Asked: every paper 2016–2026 had the same shape: Q1 any 5 of 6 × 3 = 15; Q2 any 5 of 6 × 7 = 35; Q3 and Q4 any 1 of 2 × 15 = 30 (long answers). The printed syllabus scheme (20 MCQs, 5-mark questions) was never used.

Time plan

Read and choose	Tick choices, plan both long answers: 5 min
Q1: 5 × 3	~6 min each (30 min)
Q2: 5 × 7	~14 min each (70 min)
Q3, Q4: 2 × 15	~30 min each (60 min)
Revise	Underline keywords; add doses, measurements, prevention: 15 min

~2 minutes a mark. Use the long answer's **break-up** (5 + 5 + 5, 3 + 5 + 7) as headings, space in proportion. Choose the case whose setting you know best (worker, woman, elder), not the shortest.

What is asked most (of 20 papers)

Work stress (p. 37)	17, with 16 long cases
Pelvic floor (p. 14)	16 (incontinence, prolapse)
Rehabilitation (p. 23)	15 (IBR vs CBR, team)
Obesity; labour and postnatal; ageing	14 each (obesity 10 long cases; labour 9 long answers)
Pregnancy; fitness; hazards; CBR	13; 12; 12; 11
Ability, disability management	10 each
Delivery, ageing theories, geriatric evaluation, disability	9 each; menopause, falls 8 each
Policies, elderly, women, amputee, disaster, industrial	3–7 each

Repeats (polish as model answers): theories of ageing (9 papers), work conditioning vs hardening (8), IBR vs CBR (6), FCE (6), levels of disability prevention (5), communication barriers in the elderly (3).

Short answers: 3 and 7 marks

3 marks (half a page): definition → 4–6 points or small table → one example or number; quick diagram only (ICIDH chain, T-score scale). "Enumerate" = list; "define" = standard definition **word for word** (WHO health, ICIDH, rehabilitation, ergonomics, CBR).

7 marks (1.5–2 pages): definition (1–2 lines) → table or numbered points with headings echoing the question → diagram → **PT role or prevention** → one line of evidence ("exercise reduces falls by 23%").

Long answers: five templates

- **Worker in pain:** job and task analysis; ICF, functional diagnosis; goals; treatment with doses; ergonomic changes; prevention; return to work.
- **Woman:** stage and body changes; ICF; goals; exercise and pelvic floor; posture and daily-task ergonomics; warning signs; education.
- **Older person:** assessment with scales; ICF; goals; strength, balance, gait; falls; home; institution or home; social support, schemes.
- **Weight and fitness:** BMI, waist, body fat, fitness tests with working; ICF; goals; **FITT** with heart-rate target; diet; precautions; review.
- **Disability and CBR:** ICF functional diagnosis; local resources; plan by CBR matrix (p. 26); aids, transport, concessions; referral, follow-up.

Always: ICF (six boxes), one-line functional diagnosis, short- and long-term goals with numbers, doses, a **community element** (ergonomics, home, schemes, prevention). Theory long answers: break-up as headings, diagram, table, PT role.

Diagrams and common mistakes

Draw: ICF for the case; workstation, hierarchy of controls, worker care spectrum; pelvic floor sling, diastasis test, stages of labour; fun-to-failure, balance system, ageing-theories tree; rehab team, CBR matrix, ICIDH chain; three tiers of care, disaster cycle, START; apple and pear, energy systems, circuit. Pencil, labelled, titled, referred to in the text.

Mistakes: no community part; "strengthening" without exercise, dose, progression; ignoring the break-up; missing measurements (BMI, waist, TUG, PERFECT, REBA); **old facts** (PWD 1995 replaced by RPWD 2016: 21 disabilities, 4% jobs; Asian Indian BMI cut-offs); no red flags (pre-eclampsia, osteoporotic fracture, postural hypotension, hypoglycaemia); too long on 3-mark answers.

The last week

Days 1–2: work stress, hazards, ability and disability management + one worker case in 30 min. **3–4:** women's health + one case. **5:** geriatrics + one case. **6:** health promotion and rehabilitation pages. **7:** redraw every diagram; outline the repeated 7-mark answers.

The Community practical exam: long case, synopsis and journal

Long case 50, synopsis 25, journal 5.

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Asked: not a theory question. The practical is **80 marks**: one long case (50) from rehabilitation, women's health, geriatrics, industrial health or health promotion; project synopsis (25); journal (5). Internal assessment adds 20.

Marking scheme

Long case 50	Assessment, ICF diagnosis, goals, plan, demonstration, viva
Synopsis 25	Defend the internship study: introduction, aims, methods, review of literature
Journal 5	Five cases (one per area) + at least one visit: industry, geriatric home or community assessment

Terminal: two long cases × 40. **Preliminary:** one long case 50 + synopsis 30.

Community long-case proforma

Subjective: age, sex, education, marital status, **family** (joint or nuclear, who helps), income, socioeconomic status; complaints in the person's words; history (obstetric and menstrual in women, **falls** in the elderly); **occupation** (hours, shifts, postures, loads, tools, breaks, environment; job and task analysis); **home** (floor level, stairs, **Indian or western toilet**, floor, lighting, distance to services, transport); lifestyle (activity, diet, sleep, tobacco, stress); the person's goals; **knowledge of schemes**.

Objective: vitals, height, weight, BMI, waist; posture, range, strength, flexibility, special tests; ADL and IADL, gait, balance (TUG, Berg), endurance (6-minute walk), fitness tests. **Area-specific:** PERFECT and diastasis recti (women); MMSE, GDS, frailty (elderly); REBA, RULA, FCE items (workers); disability percentage and aids (rehabilitation). **Outcome measures** to repeat: NPRS, NDI or Oswestry, ICIQ, FES-I, Barthel.

Then: ICF table, one-line functional diagnosis → problem list → short- and long-term goals → plan, home programme → community and workplace changes → schemes and referrals.

Add by area

Industrial	Job and task analysis, ergonomic score, workstation drawing, hazards, work conditioning and hardening, return to work (disability management (p. 33))
Women's health	Obstetric history, stage of pregnancy or postpartum, pelvic floor and diastasis, feeding and lifting posture, safe exercise, warning signs
Geriatrics	Comprehensive scales, falls risk, home hazards, carer, institution or home
Health promotion	Fitness tests, BMI and body composition, risk factors, **FITT with heart-rate target**, lifestyle plan
Rehabilitation	Disability type and percentage, aids, transport, accessibility, **CBR plan**, schemes, concessions, vocational plans

Presenting the case (5 minutes)

1. **One sentence:** "Mrs A, 38, homemaker with two children, leaks urine on coughing and lifting for 2 years."
2. Key findings **in ICF order** (draw the ICF if asked) → one-line functional diagnosis.
3. Two or three priority problems with short- and long-term goals.
4. Plan: treatment with doses, home programme, ergonomic or home changes, health promotion, schemes, referrals.
5. **Demonstrate one test and one treatment:** PERFECT and pelvic floor teaching; REBA and a workstation exercise; TUG and a balance exercise.

Synopsis (25) and journal (5)

Synopsis: two copies, consent form, proforma. Explain question (FINER, PICO), design, sample size, criteria, outcome measures, statistics, ethics; admit limitations (synopsis (p. 39)).

Journal: five cases with assessment, ICF diagnosis, goals, intervention, outcome. **Visit record:** date, place, purpose, people met, observations (tasks and hazards, ergonomic problems, facilities for elders, accessibility), PT recommendations, what you learnt. Neat, signed, up to date; any entry may be asked.

What examiners watch

Introduction, consent, explanation, **privacy and draping** (women's health); **safety**: BP and warning signs, guard the elderly, no heavy testing in pregnancy or osteoporosis; practical, **affordable advice** for the person's home, job, income; correct, current facts on schemes and laws; honest answers ("I would refer to the doctor"), not guesses.

Last two weeks

Practise the proforma on classmates until it takes **under 20 minutes**; rehearse one case from **each of the five areas** aloud in ICF order; rehearse the synopsis and common examiner questions; check the journal and visit report.

Abbreviations

1RM: One-repetition maximum
6MWT: Six-minute walk test
AC: Air-conditioned
ACBT: Active cycle of breathing technique
ACOG: American College of Obstetricians and Gynecologists
ACSM: American College of Sports Medicine
ADA: American Diabetes Association
ADAPT: Able Disabled All People Together (formerly the Spastics Society of India)
ADIP: Assistance to Disabled Persons for purchase of aids and appliances
ADL: Activities of daily living
AF: Atrial fibrillation
AIIMS: All India Institute of Medical Sciences
ANM: Auxiliary nurse midwife
ANOVA: Analysis of variance
ASHA: Accredited Social Health Activist
ASIA: American Spinal Injury Association
ATP: Adenosine triphosphate
BK: Below-knee
BMI: Body mass index
BMR: Basal metabolic rate
BMVSS: Bhagwan Mahaveer Viklang Sahayata Samiti (Jaipur foot)
BP: Blood pressure
BPL: Below poverty line
CBR: Community-based rehabilitation
CHC: Community health centre
CHD: Coronary heart disease
CO: Carbon monoxide (also cardiac output)
COPD: Chronic obstructive pulmonary disease
CP: Cerebral palsy
CSR: Corporate social responsibility
CT: Computed tomography
CTD: Cumulative trauma disorder
CTRI: Clinical Trials Registry–India
CTS: Carpal tunnel syndrome
DDMA: District Disaster Management Authority
DDRC: District Disability Rehabilitation Centre
DEIC: District Early Intervention Centre
DEXA: Dual-energy X-ray absorptiometry
DM: Diabetes mellitus
DNA: Deoxyribonucleic acid
DVT: Deep vein thrombosis
ECT: Every contraction timed
EMG: Electromyography
ESI: Employees' State Insurance
ESIC: Employees' State Insurance Corporation
EU: European Union
FCE: Functional capacity evaluation
FES-I: Falls Efficacy Scale–International
FIM: Functional Independence Measure
FINER: Feasible, interesting, novel, ethical, relevant (a good research question)
FITT: Frequency, intensity, time and type
FITT-VP: Frequency, intensity, time, type, volume and progression
FRC: Functional residual capacity
FSH: Follicle-stimulating hormone
FVC: Forced vital capacity
GDS: Geriatric Depression Scale
GMFCS: Gross Motor Function Classification System
HAVS: Hand–arm vibration syndrome
HR: Heart rate
HRR: Heart-rate reserve
IADL: Instrumental activities of daily living
IAP: Indian Academy of Pediatrics
IARC: International Agency for Research on Cancer
IBR: Institution-based rehabilitation
ICDS: Integrated Child Development Services
ICF: International Classification of Functioning, Disability and Health
ICIDH: International Classification of Impairments, Disabilities and Handicaps (WHO, 1980)
ICIQ: International Consultation on Incontinence Questionnaire
ICIQ-UI: ICIQ urinary incontinence short form
ICRC: International Committee of the Red Cross
ICT: Information and communication technology
ICU: Intensive care unit
IEA: International Ergonomics Association
IFT: Interferential therapy
IGNOAPS: Indira Gandhi National Old Age Pension Scheme
ILO: International Labour Organization
IP: Interphalangeal
IQ: Intelligence quotient
IT: Information technology
JSSK: Janani Shishu Suraksha Karyakram
JSY: Janani Suraksha Yojana
LAQ: Long-answer question
LH: Luteinising hormone
LPG: Liquefied petroleum gas (cooking gas)
LV: Left ventricle
MCP: Metacarpophalangeal
MET: Metabolic equivalent
METs: Metabolic equivalents (1 MET = resting energy use)
MMSE: Mini-Mental State Examination
MNA: Mini Nutritional Assessment
MRC: Medical Research Council
MRI: Magnetic resonance imaging
MS: Mitral stenosis (also multiple sclerosis)
MSDs: Musculoskeletal disorders
NCD: Non-communicable disease
NDFDC: National Divyangjan Finance and Development Corporation
NDI: Neck Disability Index
NDMA: National Disaster Management Authority
NDRF: National Disaster Response Force
NFHS-5: National Family Health Survey, fifth round (2019–21)
NGO: Non-governmental organisation
NICE: National Institute for Health and Care Excellence (UK)
NIOSH: National Institute for Occupational Safety and Health (USA)
NLEP: National Leprosy Eradication Programme
NPCBVI: National Programme for Control of Blindness and Visual Impairment
NPHCE: National Programme for Health Care of the Elderly
NPPCD: National Programme for Prevention and Control of Deafness
NPRS: Numeric Pain Rating Scale
OA: Osteoarthritis
OCRA: Occupational Repetitive Actions index
OT: Occupational therapy
OWAS: Ovako Working Posture Analysis System
$PaCO_2$: Arterial carbon dioxide pressure
PaO_2: Arterial oxygen pressure
PC: Phosphocreatine (creatine phosphate)
PCOD: Polycystic ovarian disease
PCOS: Polycystic ovary syndrome
PCPNDT: Pre-Conception and Pre-Natal Diagnostic Techniques (Act)
PE: Pulmonary embolism
PERFECT: Power, endurance, repetitions, fast contractions, every contraction timed (pelvic floor grading)
PFMT: Pelvic floor muscle training
PHC: Primary health centre
PICO: Population, intervention, comparison, outcome
PM-JAY: Pradhan Mantri Jan Arogya Yojana
PMF: Progressive massive fibrosis
PMMVY: Pradhan Mantri Matru Vandana Yojana
PMSMA: Pradhan Mantri Surakshit Matritva Abhiyan
POP-Q: Pelvic Organ Prolapse Quantification
PPE: Personal protective equipment
PT: Physiotherapy
PTB: Patellar tendon bearing
PWD: Persons with disabilities
QEC: Quick Exposure Check
RA: Rheumatoid arthritis
RBSK: Rashtriya Bal Swasthya Karyakram
RCT: Randomised controlled trial
REBA: Rapid Entire Body Assessment
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RPWD: Rights of Persons with Disabilities
RR: Respiratory rate
RULA: Rapid Upper Limb Assessment
RV: Residual volume
SACH: Solid ankle cushion heel
SCI: Spinal cord injury
SDMA: State Disaster Management Authority
SpO_2: Oxygen saturation by pulse oximeter
SPPB: Short Physical Performance Battery
SSI: Spastics Society of India (now ADAPT)
START: Simple Triage and Rapid Treatment
SUI: Stress urinary incontinence
TB: Tuberculosis
TENS: Transcutaneous electrical nerve stimulation
TLC: Total lung capacity
TUG: Timed Up and Go test
UDID: Unique Disability ID
UIP: Universal Immunisation Programme
UN: United Nations
UNESCO: United Nations Educational, Scientific and Cultural Organization
UTI: Urinary tract infection
UUI: Urge urinary incontinence
UV: Ultraviolet
VAS: Visual analogue scale
VC: Vital capacity
VO_2max: Maximal oxygen uptake
WHO: World Health Organization
WHR: Waist–hip ratio
YMCA: Young Men's Christian Association